ONLINE APPENDIX: THE COST OF VOLUNTARY EXCHANGE: ALLOCATION AND SETTLEMENT UNDER SEPARATE PUBLIC ACCOUNTS

Supplementary proofs and extensions for the accompanying main manuscript. Cross-document references refer to the revised files.

APPENDIX A: GENERAL RESERVE BOUNDS AND EXACT RECOVERY

Both results retain $B \equiv 0$, complete information, positive finite action caps, linear resource effects, divisible irreversible adjustments, complete bilateral access, demander-only transactions, and enforceable compensation balanced within each approval. The second result additionally requires full reserve repayment.

A.1. *Bounds on the approval count*

Initially let employee benefits and costs be linear, with $v_i, c_j > 0$, and put $\bar{v} = \max_i v_i$. A pair increment (x, y) is voluntary exactly when $v_i x \geq c_j y$. Let $R^0 \geq 0$, $Z = \{k : R^0_k = 0\}$, and $R^\epsilon = R^0 + \epsilon r$, where $r \geq 0$ and $r_k > 0$ for every $k \in Z$. Assume a blocked start: no nonzero voluntary approval is feasible at R^0, although $V^S(R^0) > 0$. Define the compact normalized direction set

$$\mathcal{D} = \{e_i : i \in D\} \cup \bigcup_{i,j} \{x e_i + y e_j : x, y \geq 0,\ x + y = 1,\ v_i x \geq c_j y\}.$$

For any $\lambda \gg 0$ on the initially empty accounts, put

$$\delta_\lambda = \min_{d \in \mathcal{D}} \sum_{k \in Z} \lambda_k [-(Md)_k]_+, \qquad g_\lambda = \max\{0, \max_{d \in \mathcal{D}} \lambda^\mathsf{T} M_Z d\}, \qquad S_\lambda = \lambda^\mathsf{T} r_Z. \quad \text{(A.1)}$$

These analytical weights do not permit transfers between accounts.

THEOREM A.1—General approval bounds: *Under these assumptions, $\delta_\lambda, S_\lambda > 0$, and for every $\epsilon > 0$ and integer $T \geq 0$,*

$$V_T^{\mathrm{seq}}(R^\epsilon) \leq \begin{cases} \dfrac{\bar{v}\epsilon S_\lambda}{g_\lambda}[(1 + g_\lambda/\delta_\lambda)^T - 1], & g_\lambda > 0, \\ \dfrac{\bar{v}\epsilon S_\lambda}{\delta_\lambda} T, & g_\lambda = 0. \end{cases} \quad \text{(A.2)}$$

Fix a planner optimum z^* at R^0 and $0 < \eta < V^S(R^0)$. The target $z^\eta = qz^*$, where $q = \eta/V^S(R^0)$, admits exact voluntary implementation in at most

$$B_\eta/\epsilon + C_\eta \tag{A.3}$$

approvals, for finite computable constants independent of ϵ. If

$$\lambda^{\mathsf{T}} M_Z d \le 0 \quad (d \in \mathcal{D}), \tag{A.4}$$

then, writing T_η for the minimum approvals achieving surplus at least η and N_η for those implementing z^η,

$$\frac{\eta\delta_\lambda}{\bar{v}S_\lambda\epsilon} \le T_\eta(\epsilon) \le N_\eta(\epsilon) \le \frac{B_\eta}{\epsilon} + C_\eta. \tag{A.5}$$

Thus both requirements have inverse-reserve order under the certificate.

PROOF: If $\delta_\lambda = 0$, compactness gives $d \in \mathcal{D}$ with $M_Z d \ge 0$. A sufficiently small positive multiple respects every positive initial account and action cap, contradicting the blocked start. That assumption also implies $Z \neq \varnothing$, so $S_\lambda > 0$. Write an approval as $s_t d_t$, $d_t \in \mathcal{D}$, and let $H_t = \lambda^{\mathsf{T}}(R^\epsilon + Mz_t)_Z$. Accountwise feasibility implies

$$\delta_\lambda s_t \le H_{t-1}, \qquad H_t \le H_{t-1} + g_\lambda s_t, \qquad H_0 = \epsilon S_\lambda.$$

Iterating and summing these inequalities, then using $W(z_T) \le \bar{v}\sum_{t\le T} s_t$, proves (A.2).

For the upper bound, use the decomposition proved below in Theorem A.2. Its marginal-financing condition holds for the linear target because $Q - C = W(z^\eta) = \eta > 0$. This gives voluntary bundles $u^1, \dots, u^L$ summing to z^η, where $L \le |D||S| + |D|$. Set

$$A_k = \sum_\ell [-(Mu^\ell)_k]_+, \qquad a_\eta = \max_{k\in Z} A_k/r_k, \qquad b_\eta = \max_{k\notin Z} \frac{A_k}{(1-q)R_k^0},$$

with an empty maximum equal to zero, and choose

$$n = \max\left\{1, \left\lceil \max_k \frac{A_k}{(1-q)R_k^0 + \epsilon r_k} \right\rceil \right\}. \tag{A.6}$$

All denominators are positive. Execute n identical ordered rounds of $u^1/n, \dots, u^L/n$. A round starting at fraction t of z^η has balances $(1-tq)R^0 + tq(R^0 + Mz^*) + \epsilon r \ge (1 -$

$q)R^0+\epsilon r$. Each within-round decrease is at most A_k/n, so every prefix is feasible. Activities increase to the target and every divided bundle remains voluntary. The rounds follow Fraca and Haddad (2015, Lemma 13); the decomposition verifies compensation.

Since $n \le a_\eta/\epsilon + 1 + b_\eta$, take $B_\eta = La_\eta$ and $C_\eta = L(1+b_\eta)$. Finally, (A.4) implies $g_\lambda = 0$; inserting the required surplus η in (A.2) gives (A.5). *Q.E.D.*

A.2. *Exact recovery with repayment*

THEOREM A.2—Exact recovery with a repayable reserve: *Retain the section's institutional assumptions. Let $R^0 \ge 0$ and $r \gg 0$. Benefits are differentiable, concave, and nondecreasing; costs are differentiable and convex with strictly positive derivatives, with $U_i(0) = C_j(0) = 0$. Fix any original-ledger planner optimum $z^* = (a^*, b^*)$ with $V_0 = W(z^*) = V^S(R^0) > 0$. For every $\epsilon > 0$, a finite sequence of voluntary approvals with balanced employee compensation implements z^* from $R^0 + \epsilon r$ and returns the full reserve. Hence the original planner value is exactly recoverable before implementation costs.*

There is a decomposition $z^* = \sum_{\ell=1}^{L} u^\ell$ into nonzero bilateral or demander-only bundles, with $L \le |D||S| + |D|$. Define $A_k^* = \sum_\ell [-(Mu^\ell)_k]_+$ and $H^* = \max_k A_k^*/r_k$. The minimum approvals implementing this optimum with repayment satisfy

$$N^*(\epsilon) \le L \max\{1, \lceil H^*/\epsilon \rceil\} \le LH^*/\epsilon + L.$$

With additionally positive linear preferences, a blocked start, and (A.4)*, also $N^*(\epsilon) \ge V_0\delta_\lambda/(\bar{v}S_\lambda\epsilon)$, using* (A.1)*. The inverse-reserve order is therefore necessary and sufficient in that class.*

PROOF: Let $g_i = U_i'(a_i^*) \ge 0$, $h_j = C_j'(b_j^*) > 0$, $Q = \sum_i g_i a_i^*$ and $C = \sum_j h_j b_j^*$. The segment from 0 to z^* is feasible. Optimality at its endpoint gives the inward radial condition $Q \ge C$. If $C > 0$, also $Q > 0$, and define pair increments

$$x_{ij} = a_i^* h_j b_j^*/C, \qquad y_{ij} = b_j^* g_i a_i^*/Q$$

for each demander and active supplier. They satisfy $\sum_j x_{ij} = a_i^*$, $\sum_i y_{ij} = b_j^*$, and $g_i x_{ij} \ge h_j y_{ij}$. When $g_i = 0$ the bundle is demander-only. If $C = 0$, strictly positive h_j implies $b^* = 0$,

and one demander-only bundle per active demander suffices. Discarding zero bundles gives the stated u^ℓ and bound on L.

Execute $n = \max\{1, \lceil H^*/\epsilon \rceil\}$ rounds of $u^1/n, \ldots, u^L/n$. At fraction t of completed rounds, balances are $(1-t)R^0 + t(R^0 + Mz^*) + \epsilon r \geq \epsilon r$. The greatest within-round decrease is at most $A_k^*/n \leq \epsilon r_k$ in each account. All intermediate balances are nonnegative, and activities increase coordinatewise from zero to z^*, respecting caps. Linear resource effects preserve feasibility within each approval.

At any such cumulative allocation, concavity and convexity imply

$$\Delta U_i \geq g_i x_{ij}/n \geq h_j y_{ij}/n \geq \Delta C_j.$$

Charge the demander exactly the supplier's incremental cost and pay it to that supplier. Compensation balances and both weakly gain. Demander-only increments have nonnegative benefits and need no transfer. Thus participation holds at each approval under nonlinear preferences.

The final balances are $R^0 + Mz^* + \epsilon r$, so the reserve is fully repayable. Conversely, repayment at any terminal allocation z implies $R^0 + Mz \geq 0$ and therefore $W(z) \leq V^S(R^0)$. There are $Ln \leq LH^*/\epsilon + L$ approvals. Under the additional linear, blocked-start, and nonexpansion assumptions, (A.2) with surplus V_0 and $g_\lambda = 0$ supplies the positive matching lower bound. *Q.E.D.*

APPENDIX B: EXACT RESERVE AND COST FRONTIERS

B.1. *Proof of the exact productive frontier*

PROOF OF PROPOSITION 2: First take $h > 0$ and abbreviate $\gamma = \gamma_h$ and $\epsilon = e$. At any cumulative allocation, write P, W, H for the three balances and set

$$S = P + W + H = 3e + hb_1, \qquad q = W - H = \frac{b_1 - b_2}{2},$$

$$u = S - q = P + 2H, \quad v = S + q = P + 2W.$$

Feasibility implies $u, v \geq 0$, with initial state $(3e, 3e)$. A supplier-1 increment y changes (u, v) by $(-(1/2 - h)y, (1/2 + h)y)$: it converts some of u into v at rate γ. A supplier-

2 increment changes (u, v) by $(y/2, -y/2)$ and converts some of v into u at unit rate. Demander-only transactions leave both coordinates unchanged. The target requires

$$u = v = s, \qquad s = 3e + h\theta/2.$$

These two-coordinate restrictions give a lower bound for every physical schedule. In this relaxed problem, delete transactions with no supplier increment and combine adjacent increments of the same supplier. This deletion is not asserted to preserve the original physical balances. The remaining word alternates and has length at most T; it has at least two steps because the target uses both suppliers. To bound its final common stock, work backward. If its last transaction uses supplier 1, equality of the terminal coordinates gives $s = (\gamma u + v)/(1 + \gamma)$ at the preceding state. If it uses supplier 2, it gives $s = (u + v)/2$. For a nonnegative linear bound $Au + Bv$, a preceding supplier-1 conversion changes the coefficient on u to $\max\{A, \gamma B\}$; a preceding supplier-2 conversion changes the coefficient on v to $\max\{B, A\}$. These replacements follow by maximizing the converted amount over $[0, u]$ and $[0, v]$, respectively, so they bound every admissible partial conversion.

For an alternating word, the replacements select full conversion at every prefinal step. Evaluated at $(3e, 3e)$, a word beginning with supplier 1 and of length $2n$ has bound $3e(1 + \gamma)\gamma^{n-1}/2$; one beginning with supplier 2 has bound $3e\,2\gamma^n/(1 + \gamma)$. The first is weakly larger, because $(1 + \gamma)^2 \geq 4\gamma$. At length $2n + 1$, both bounds are $3e\gamma^n$. Thus $s \leq 3eG_T(h)$ for length T. The sequence $G_T(h)$ is strictly increasing, so the same bound covers all shorter words. Substituting the required s gives $e \geq h\theta/[6(G_T(h) - 1)]$.

It remains to lift an attaining conversion path to physical, voluntary transactions. Choose e equal to this lower bound, begin with supplier 1, and alternate suppliers. Make every prefinal conversion full. For a prefinal supplier-1 step, select the demander increment so that its endpoint has $P = H = 0$; equivalently, its new cumulative demander quantity is $a = b_1 + b_2 - e$. For a prefinal supplier-2 step, select the increment so that $P = W = 0$; equivalently, set $a = b_1/2 + 3b_2/2 - e$. The other account is nonnegative in each case. The first step has $y = 6e/(1 - 2h)$ and demander increment $x = y - e > 0$. An interior supplier-1 step has $x = (3 + 2h)y/4$, and an interior supplier-2 step has $x = 5y/4$.

Use the last conversion to equalize u and v. The amplification calculation gives $u = v = 3eG_T(h) = 3e + h\theta/2$. The identities for S and q therefore give $b_1 = b_2 = \theta/2$. Set the final

demander quantity to $a = \theta$. If the last supplier is 1, its increments satisfy $x = y/2 + e$; if it is 2, they satisfy $x = y + e$. The resulting final balances are $(e + h\theta/2, e, e)$, so the full reserve can be returned.

Every displayed increment is nonnegative and satisfies $x > y/10$. Paying the supplier $y/10$ and charging the demander the same amount makes each transaction weakly voluntary and exactly balanced. Since all coordinates increase to the target, every prefix respects the unit action caps. The prefinal endpoints and the final endpoint are resource-feasible; linear account effects also preserve feasibility along each divisible transaction. Any larger reserve permits the same schedule with the excess capacity retained throughout.

At $h = 0$, the conserving-economy proof gives $e = \theta/[12(T-1)]$. Finally, $\gamma_h = 1 + 4h + O(h^2)$ and $G_T(h) = 1 + 2h(T-1) + O(h^2)$ for fixed T, which gives the asserted continuous limit. *Q.E.D.*

B.2. *The conserving frontier*

PROOF OF (B.1): Fix $\theta \in (0, 1]$ and a schedule with at most T bilateral settlements that implements $z_\theta = (\theta, \theta/2, \theta/2)$ from $\epsilon(1, 1, 1)$. Zero increments can be appended to give exactly T steps. Since both suppliers must make positive adjustments, $T \geq 2$. Write the resource balances after step t as (P_t, W_t, H_t) and put $q_t = W_t - H_t$. Every resource column sums to zero, so feasibility gives

$$P_t + W_t + H_t = 3\epsilon, \qquad -3\epsilon \leq q_t \leq 3\epsilon.$$

The target has zero effect on every resource account; hence $q_0 = q_T = 0$. A supplier-1 transaction of size (x, y) changes q_t by $y/2$, while a supplier-2 transaction changes it by $-y/2$. The demander's increment cancels from $W_t - H_t$. Therefore

$$\theta = b_1 + b_2 = 2\sum_{t=1}^{T} |q_t - q_{t-1}| \leq 2\{3\epsilon + 6\epsilon(T-2) + 3\epsilon\} = 12\epsilon(T-1).$$

This proves the lower bound for every voluntary schedule, regardless of the supplier order or the division of the demander's adjustment.

For attainment, set $e = \theta/[12(T-1)]$. Start with a supplier-1 transaction $(x, y) = (5e, 6e)$. It changes the resource balance from (e, e, e) to $(0, 3e, 0)$. Each interior transaction alternates

suppliers: supplier 2 uses $(15e, 12e)$ and changes $(0, 3e, 0)$ to $(0, 0, 3e)$; supplier 1 uses $(9e, 12e)$ and reverses that change. After $T-2$ interior transactions, finish with supplier 1 and $(4e, 6e)$ if T is odd, or with supplier 2 and $(7e, 6e)$ if T is even. Either final transaction returns the balance to (e, e, e).

The first and last increments of q_t have absolute size $3e$, and every interior increment has absolute size $6e$. Hence $b_1 + b_2 = 12e(T-1) = \theta$. Since $q_T = q_0$, the two suppliers' totals are equal, giving $b_1 = b_2 = \theta/2$. The unchanged final position balance then gives $a = b_1 + b_2 = \theta$. All increments are nonnegative, so the target respects every individual cap and so does each prefix. Each displayed transaction satisfies $x - y/10 > 0$ and therefore admits balanced compensation with both employees weakly better off. Every prefix has nonnegative public balances. Thus the lower bound is attained for every $\theta \in (0, 1]$ and integer $T \geq 2$. Any larger reserve also permits the same schedule, completing the proof. *Q.E.D.*

B.3. *Exact costs and the fixed-target comparison*

PROOF OF THEOREM 11: An admissible T-approval schedule requires reserve at least ϵ_T, and Proposition 2attains this amount. Taking infima proves (5). The minimum exists because the cost is at least τT. A supplier-1 pair $(\theta/2, \theta/2)$ followed by the same supplier-2 pair is voluntary and feasible from $\theta/4$ in each account, and returns the reserve. No schedule uses fewer than two approvals, so any larger reserve is dominated.

Under conservation, writing $n = T - 1$ gives the scalar objective $\rho\theta/(12n) + \tau(n + 1)$; the arithmetic–geometric mean inequality and rounding its real minimizer establish the stated order. For fixed $h > 0$, the formula for G_T gives constants $A_1, A_2, b > 0$ with $A_1 \exp(-bT) \leq \epsilon_T \leq A_2 \exp(-bT)$ for every $T \geq 2$. Choose $T = \lceil b^{-1} \log(\rho/\tau) \rceil + 2$ for an upper bound. For the lower bound, if $T \geq (2b)^{-1} \log(\rho/\tau)$ the approval cost already has the required order. Otherwise the reserve cost is at least $A_1\sqrt{\rho\tau}$, which exceeds a constant multiple of $\tau \log(\rho/\tau)$ as $\tau/\rho \downarrow 0$. This proves both bounds. *Q.E.D.*

COROLLARY A.3—A strict reversal of the preferred settlement rule: *Fix $\theta = 1$, $\rho = 12$, $\tau = 1/10000$, $F = 1/100$, and $\mathcal{K} = 0$. At $h = 0$, the sequential minimum is attained at reserve $1/1200$ with 101 approvals, and*

$$Q_0 = \frac{201}{10000} > \frac{101}{10000} = \tau + F.$$

At $h = 1/4$, the sequential minimum is attained at reserve $1/104952$ with 16 approvals, and

$$Q_{1/4} = \frac{9371}{5466250} < \frac{101}{10000} = \tau + F.$$

Joint settlement is therefore strictly cheaper in the conserving economy, while sequential settlement is strictly cheaper after the productive perturbation. In each economy, implementing the target is strictly better than retaining the baseline under the stated cost criterion.

PROOF OF COROLLARY A.3: The exact conserving threshold (B.1) gives

$$Q_0 = \tau + \min_{n \geq 1} \left\{ \frac{\rho}{12n} + \tau n \right\}.$$

For the specified prices, the expression in braces is $1/n + n/10000$. Its unique real minimizer is $n = 100$, which is an integer. The minimum is $1/50$, giving $Q_0 = 201/10000$, 101 approvals, and reserve $1/1200$.

For $h = 1/4$, the exact threshold in (4)gives the cost for T approvals as

$$c_T = \frac{1}{2(g_T - 1)} + \frac{T}{10000}, \qquad g_{2n} = 2 \cdot 3^{n-1}, \quad g_{2n+1} = 3^n.$$

Within either parity, increasing T by two multiplies g_T by three. The corresponding cost difference is

$$\frac{1}{5000} - \frac{g_T}{(3g_T - 1)(g_T - 1)}.$$

These differences strictly increase as g_T increases, because $g/[(3g-1)(g-1)]$ is strictly decreasing for $g \geq 2$. For the even subsequence the difference is negative at $g_{14} = 1458$ and positive at $g_{16} = 4374$, so its unique minimum is at $T = 16$. For the odd subsequence it is negative at $g_{13} = 729$ and positive at $g_{15} = 2187$, so its unique minimum is at $T = 15$. Finally,

$$c_{15} - c_{16} = \frac{687811}{47796890000} > 0.$$

Hence $T = 16$ is the global minimizer. Its threshold reserve is $1/[24(4374-1)] = 1/104952$, and its cost is $1/8746 + 16/10000 = 9371/5466250$. The common joint cost is $101/10000$, strictly between the two exact sequential minima. Both preferred procedures cost less than the target surplus $9/10$, and $\mathcal{K} = 0$, establishing adoption as well. *Q.E.D.*

B.4. *Endogenous choice in the conserving economy*

The preceding theorem solves the conserving allocation choice at the stated prices. The same argument yields a complete comparative-static solution for M_0. Repayment requires $M_0 z \geq 0$; zero column sums force equality. Thus every terminal allocation is z_θ, $0 \leq \theta \leq 1$. The frontier specializes to

$$\epsilon_T(\theta) = \frac{\theta}{12(T-1)}, \qquad T \geq 2. \tag{B.1}$$

Allow common verification cost $\mathcal{K}(z_\theta) = k\theta$, $k \geq 0$, and retain the same fixed $F \geq 0$ for every nonzero joint package. The baseline incurs no cost. If a reserve cap is imposed, assume it is at least $1/12$, so it does not restrict any minimum reserve in this comparison.

PROPOSITION A.4—Endogenous quantity, reserve, and approval rule: *Under these conditions, define*

$$G(\rho, \tau) = \min_{n \in \{1,2,\ldots\}} \left\{ \frac{\rho}{12n} + \tau n \right\}. \tag{B.2}$$

The authority's maximum net value, allowing either institution and the baseline, is

$$\mathcal{V}(\rho, \tau, F, k) = \max \left\{ 0, \frac{9}{10} - k - \tau - \min\{F, G(\rho, \tau)\} \right\}. \tag{B.3}$$

Whenever positive net surplus is attainable, an optimal allocation is $z^* = (1, 1/2, 1/2)$. Joint approval implements it at lower cost when $F < G(\rho, \tau)$, and sequential approval does so at lower cost when $F > G(\rho, \tau)$. Under sequential approval, an integer $n^* \geq 1$ minimizes (B.2) *if and only if*

$$\tau n^*(n^* - 1) \leq \frac{\rho}{12} \leq \tau n^*(n^* + 1). \tag{B.4}$$

Conditional on implementing z^* sequentially, the least-cost choice uses $n^* + 1$ approvals and reserve $\epsilon^* = 1/(12n^*)$.

PROOF OF PROPOSITION A.4: Fix $\theta > 0$ and $T = n + 1 \geq 2$. Since reserve is costly, its optimal amount for that pair (θ, n) is the minimum in (B.1). Sequential net value is therefore

$$\theta \left(\frac{9}{10} - k - \frac{\rho}{12n} \right) - \tau(n+1).$$

For a fixed n, if the coefficient of θ is positive, this expression is maximized at $\theta = 1$. If it is nonpositive, every positive allocation gives negative net value and the baseline is better. Joint net value is $\theta(9/10 - k) - \tau - F$ and has the same full-quantity-or-baseline property. Maximizing over n, the two institutions, and the baseline gives (B.3).

Write $f(n) = \rho/(12n) + \tau n$. Since

$$f(n+1) - f(n) = \tau - \frac{\rho}{12n(n+1)},$$

its consecutive differences increase with n. A global minimum is characterized by $f(n) - f(n-1) \leq 0 \leq f(n+1) - f(n)$ for $n \geq 2$, and just the second inequality for $n = 1$. These conditions are exactly (B.4). The minimum exists because $f(n) \to \infty$ as $n \to \infty$. ***Q.E.D.***

APPENDIX C: EXECUTION OF PRODUCTIVE RELEASES

PROOF OF THEOREM 4: Use the objects in Definition 3and set $s_0 = \sigma\epsilon$, so $\epsilon r \geq s_0 p$. After productive bundles of cumulative size $a \leq \alpha$, balances are $\epsilon r + ap$. A further repetition $tu^1, \ldots, tu^k$ is feasible whenever

$$0 < t \leq \min\{\alpha - a, (s_0 + a)/\kappa\},$$

$$\epsilon r + ap + tM \sum_{j=1}^{s} u^j \geq (s_0 + a - t\kappa)p \geq 0.$$

Its terminal balances are $\epsilon r + (a + t)p$. Choosing $t = (s_0 + a)/\kappa$ multiplies $s_0 + a$ by $\Lambda = 1 + 1/\kappa$. Consequently, maximal repetitions followed by a truncated final repetition reach $a = \alpha$ exactly in

$$n = \left\lceil \frac{\log(1 + \alpha/(\sigma\epsilon))}{\log \Lambda} \right\rceil$$

repetitions. The completion list then has prefix balances $\epsilon r + \alpha p + M \sum_{j=1}^{s} v^j \geq \epsilon r$ and terminates at $z^* = \alpha d + c$. Returning the reserve leaves $Mz^* \geq 0$. Thus the total number of approvals is at most $kn + \ell$.

All increments are nonnegative and sum to z^*, so every partial allocation respects its caps. Linear interpolation preserves account feasibility within each approval. For a bilateral increment (x, y) involving demander i and supplier j, transfer $c_j y$ from the demander to

the supplier: the gains are $v_i x - c_j y \geq 0$ and zero, respectively. Demander-only increments need no transfer. Compensation is therefore voluntary and exactly balanced.

For the lower bound, fix any $\lambda \gg 0$. Blocked start and compactness give $\delta_\lambda > 0$ in (A.1). Also $g_\lambda > 0$: otherwise $\lambda^\mathsf{T} M u^j \leq 0$ for every bundle, contradicting $\lambda^\mathsf{T} \sum_j M u^j = \lambda^\mathsf{T} p > 0$. Theorem A.1 implies

$$T \geq \frac{\log(1 + g_\lambda W(z^*)/(\bar{v}\epsilon S_\lambda))}{\log(1 + g_\lambda/\delta_\lambda)} = \Omega(\log(1/\epsilon)).$$

This matches the upper order. With a nonexpansion certificate instead, $g_\lambda = 0$ and the same theorem gives $T \geq W(z^*)\delta_\lambda/(\bar{v} S_\lambda \epsilon)$. For a planner optimum, Theorem A.2 supplies a repayable $O(1/\epsilon)$ schedule under the maintained complete bilateral access. This proves the contrasting inverse-reserve order. The summation argument above also proves incompatibility of the two certificates; it does not make either one necessary for all technologies. *Q.E.D.*

Membership of M_h. For $0 < h < 1/2$, choose

$$u^1 = (1,1,0)^\mathsf{T}, \quad u^2 = (1,0,1)^\mathsf{T},$$

$$d = (2,1,1)^\mathsf{T}, \quad p = (h,0,0)^\mathsf{T}, \quad \kappa = \frac{1/2 - h}{h}.$$

The bundles each have surplus $9/10$, and their resource changes are $(h - 1/2, 1/2, 0)^\mathsf{T}$ and $(1/2, -1/2, 0)^\mathsf{T}$. Thus every prefix is at least $-\kappa p$. Take

$$\alpha = \frac{1}{2+3h}, \qquad c = v^1 = \frac{(3h,0,2h)^\mathsf{T}}{2+3h}, \qquad M_h c = \frac{(-h,0,h)^\mathsf{T}}{2+3h}.$$

The completion bundle has positive surplus $14h/[5(2+3h)]$, satisfies $\alpha p + M_h c \geq 0$, and gives $\alpha d + c = z_h^*$. Only scaled bundles are executed; their nonnegative cumulative activities are bounded by z_h^* even though the auxiliary vector d exceeds a unit cap. For $r = (1,1,1)^\mathsf{T}$, $\sigma = 1/h$ and $\Lambda = 1/(1-2h)$.

The zero start is blocked throughout $0 \leq h < 1/2$. A supplier-1 pair would require $x \geq y$ and $x \leq (3/4 + h/2)y < y$; a supplier-2 pair would require $x \geq 3y/2$ and $x \leq 5y/4$. A positive demander-only move consumes pension capacity. At $h = 0$, every column sums to zero, giving the nonexpansion certificate $(1,1,1)^\mathsf{T}$. The general order comparison therefore applies to (M_0, M_h); the exact constants and parity terms in Proposition 2require the separate frontier proof.

APPENDIX D: A PRIMITIVE LP CERTIFICATE AND AN INTERACTING RELEASE CYCLE

PROOF OF PROPOSITION 5: The planner LP attains V^* on its compact activity box. For a fixed support and order, the second program is also a linear program with compact feasible set: nonnegative bundle activities sum to a cap-bounded vector, and the nonempty support bounds η through Mz. Thus any positive value $L_{R,\pi}$ is attained.

At an attaining solution, feasibility and $W(z) = V^*$ make z a planner optimum. Its release $p = Mz$ is positive on R and zero elsewhere. Every bundle is voluntary. Outside R, the LP makes every resource prefix nonnegative; inside R, the finite maximum defining κ bounds every prefix below by $-\kappa p$. If $\kappa = 0$, the first nonzero bundle would be feasible from zero, contradicting the blocked start. Such a bundle exists since $V^* > 0$. Also $\sigma > 0$ because $r \gg 0$ and $p > 0$ on R. Delete zero bundles. The remaining list satisfies Definition 3with $d = z$, $\alpha = 1$, and no completion. Theorem 4gives the stated count, matching lower bound, balanced compensation, and repayment. Its cumulative activities stay below z, hence below the caps. The assumed comparison technology then meets Corollary 8, proving the institutional claim. *Q.E.D.*

The finite enumeration searches over supports and orders. More general certificates can use repeated transaction types or a separate completion.

The original family. For $0 < h < 1/2$, write $D_h = 2 + 3h$ and choose $u^1 = D_h^{-1}(3/4 + h/2, 1, 0)^{\mathsf{T}}$ and $u^2 = D_h^{-1}((5/4)(1 + 2h), 0, 1 + 2h)^{\mathsf{T}}$. They sum to z_h^* and are voluntary: their demander-to-supplier ratios exceed $1/10$. Direct multiplication gives

$$M_h u^1 = \frac{(0, 1 + 2h, -1 + 2h)^{\mathsf{T}}}{4D_h}, \qquad M_h u^2 = \frac{(0, -1 - 2h, 1 + 2h)^{\mathsf{T}}}{4D_h}.$$

Thus $p = (0, 0, h/D_h)^{\mathsf{T}}$. The LP passes with $R = \{3\}$, supplier 1 before supplier 2, and zero demander-only activity. For uniform reserves, $\kappa = (1 - 2h)/(4h)$, $\sigma = D_h/h$, and $1 + 1/\kappa = \gamma_h$. This repeats portions of the planner optimum while amplifying position capacity; it needs no separate completion transaction and does not assert the smallest approval count.

The interacting four-account cycle. For $\widetilde{M}_h$ and the primitives in Section 4.3, account 1 requires $\sum_j b_j \geq a$. Hence $W \leq 9a/10 \leq 9/10$. The feasible allocation $(1, 1/3, 1/3, 1/3)$

attains equality. Any maximizer must have $a = 1$ and $\sum_j b_j = 1$; the caps $b_j \le 1/3$ then prove uniqueness.

At zero balances, a supplier-1 pair requires both $x \le y$ and $x \ge 2y$, so $x = y = 0$. A supplier-2 pair consumes y units of account 3, and a supplier-3 pair consumes y units of account 4; either therefore requires $y = 0$. With $y = 0$, any $x > 0$ consumes account 1. Supplier-only moves are not voluntary because costs are positive. The start is consequently blocked for every $h \ge 0$.

Let $w^j = e_a + e_{b_j}$, each with surplus $9/10$. Their effects are

$$\widetilde{M}_h w^1 = (0, -1, 1, 0)^{\mathsf{T}}, \qquad \widetilde{M}_h w^2 = (0, 0, -1, 1)^{\mathsf{T}},$$
$$\widetilde{M}_h w^3 = (0, 1 + h, 0, -1)^{\mathsf{T}}.$$

The optimum decomposes as $u^j = w^j/3$. Only account 2 has a negative prefix, of size $1/3$; accounts 3 and 4 finance successive steps. The net release is $p = (0, h/3, 0, 0)^{\mathsf{T}}$. The LP therefore passes with $R = \{2\}$ and supplier order $(1, 2, 3)$, giving $\kappa = 1/h$, $\sigma = 3/h$, and the count $3\lceil \log(1 + h/(3\epsilon))/\log(1 + h)\rceil$.

At $h = 0$, all columns sum to zero, so $\lambda = \mathbf{1}$ is a nonexpansion certificate. Normalize $x + y = 1$; voluntary pairs have $0 \le y \le 10/11$. The negative-resource sums for suppliers 1, 2, and 3 are respectively

$$[1 - 2y]_+ + [3y - 1]_+, \qquad |1 - 2y| + y, \qquad [1 - 2y]_+ + y.$$

Their minima are $1/3$, $1/2$, and $1/2$; a normalized demander-only move has sum one. Hence $\delta_\lambda = 1/3$, $S_\lambda = 4$, $\bar{v} = 1$, and $a_C = 1/12$. Since $\widetilde{M}_h \ge \widetilde{M}_0$, the general institutional comparison applies.

At $\rho = 1$, $\tau = 10^{-6}$, and $F = 1/5000$, the conserving sequential value is bounded above by $9/10 - 2\sqrt{10^{-6}(1/12)(9/10)} = 9/10 - \sqrt{3/10^7}$. The joint cost $201/10^6$ is smaller than this cost bound: squaring compares $40401/10^{12}$ with $300000/10^{12}$. Thus the exact best joint value $899799/10^6$ is higher. For $h = 1$ and $\epsilon = 10^{-6}$, $2^{18} < 1 + 1/(3\epsilon) < 2^{19}$, so 57 approvals suffice. The implementation cost is at most $58/10^6$, giving net value at least $449971/500000$, which exceeds the best joint value by $143/10^6$. Truncating the last cycle implements the optimum exactly; the general construction proves all prefix balances, caps,

compensation, and full reserve repayment. These optimized upper and attainable lower bounds establish the reversal without asserting exact sequential values.

APPENDIX E: OPTIMIZED IMPLEMENTATION LOSSES AND INSTITUTIONAL THRESHOLDS

PROOF OF THEOREM 6: Write $V = V_M > 0$. Repayment implies $0 < W \le V$ for any schedule that can improve on the baseline. All constants below concern the fixed economy and reserve direction; $\rho > 0$ is fixed while $\tau \downarrow 0$.

Nonexpansion. Fix a strictly positive certificate λ and put $a = \delta_\lambda/(\bar{v}S_\lambda) > 0$, using (A.1). The blocked-start bound gives $T \ge aW/\epsilon$, so

$$\rho\epsilon + \tau T \ge 2\sqrt{\rho\tau a W}.$$

The function $W - 2\sqrt{\rho\tau a W}$ is continuous and convex on $[0, V]$, with second derivative $\sqrt{\rho\tau a}/(2W^{3/2})$ for $W > 0$. Its maximum is therefore attained at an endpoint. Taking the supremum over allocations and schedules gives

$$\mathcal{V}_M^{\mathrm{seq}} \le \max\{0, V - 2\sqrt{\rho\tau a V}\}, \qquad L_M \ge \min\{V, 2\sqrt{\rho\tau a V}\}.$$

For the upper bound, Theorem A.2 implements a planner optimum with at most $A/\epsilon + B$ approvals, where $A = LH^* > 0$ and $B = L > 0$. Positivity of H^* follows from the blocked start: otherwise every nonzero decomposition bundle would have no negative resource effect and would itself be a feasible first move. Choosing $\epsilon = \sqrt{A\tau/\rho}$ consequently yields

$$\min\{V, 2\sqrt{\rho\tau a V}\} \le L_M \le \min\{V, 2\sqrt{\rho\tau A} + B\tau\}.$$

These bounds establish the square-root order and positive adoption for all sufficiently small τ.

Executable productivity. Use the certificate constants $k, \ell, \alpha, \sigma, \Lambda$ from Theorem 4. For any strictly positive weights λ, the blocked start gives $\delta_\lambda > 0$ and the productive release gives $g_\lambda > 0$: otherwise summing its voluntary bundles would imply $\lambda^{\mathsf{T}} p \le 0$, contradicting $p \ge 0$, $p \ne 0$. Define

$$\beta = \log(1 + g_\lambda/\delta_\lambda) > 0, \qquad c = \frac{g_\lambda V}{2\bar{v}S_\lambda} > 0.$$

An allocation with $W < V/2$ loses at least $V/2$ before costs. For $W \geq V/2$, the general blocked-start bound instead gives $T \geq \beta^{-1}\log(1 + c/\epsilon)$. With $y = \beta\rho\epsilon/\tau$ and $t = \beta\rho c/\tau$, use

$$y + \log(1 + t/y) \geq y - \log y + \log t \geq 1 + \log t$$

to bound its implementation cost below. Thus, for $0 < \tau \leq \beta\rho c$,

$$L_M \geq \min\left\{\frac{V}{2}, \frac{\tau}{\beta}\left[1 + \log\left(\frac{\beta\rho c}{\tau}\right)\right]\right\}.$$

The inequality covers the baseline and every feasible schedule, hence also the infimum of their losses; no maximizing schedule is assumed. The productive construction, applied to a planner optimum with $\epsilon = \tau/\rho$, supplies the complementary bound

$$L_M \leq \min\left\{V, \tau(1 + k + \ell) + \frac{k\tau}{\log\Lambda}\log\left(1 + \frac{\alpha\rho}{\sigma\tau}\right)\right\}.$$

Here $\lceil x \rceil \leq x + 1$ bounds the integer approval count. The two bounds have order $\tau\log(\rho/\tau)$, and the upper bound tends to zero. This proves the second rate and positive adoption. ***Q.E.D.***

PROOF OF PROPOSITION 7: Write $S = \mathcal{V}_M^{\text{seq}} = V_M - L_M$ and $J = \max\{0, V_M - \tau - F\}$. If $S > 0$, direct comparison shows $J \gtreqless S$ exactly when $F \lesseqgtr L_M - \tau$. If $S = 0$, joint settlement is strictly better exactly when $F < V_M - \tau$; otherwise both institutions choose the baseline.

A nonzero fully repayable one-approval schedule would be a voluntary bundle z with $Mz \geq 0$, contradicting the blocked start. Therefore $T \geq 2$ for every nonzero schedule and $S \leq \max\{0, V_M - 2\tau\}$. If $S > 0$, this gives $F_M^* = L_M - \tau \geq \tau$. Without a blocked start, every nonzero schedule still has $T \geq 1$. Joint settlement with $F = 0$ can implement its terminal allocation with balanced voluntary transfers, saving $\rho\epsilon + \tau(T - 1)$; the baseline covers nonpositive-surplus endpoints. Taking suprema proves weak dominance. Under the blocked start and $V_M > \tau$, the inequality $V_M - \tau > \max\{0, V_M - 2\tau\} \geq S$ proves strict dominance. ***Q.E.D.***

PROOF OF COROLLARY 8: Positive adoption holds for small τ. Subtracting τ preserves both loss orders, so

$$0 \leq \frac{F_P^*}{F_C^*} \leq C\sqrt{\frac{\tau}{\rho}}\log(\rho/\tau) \longrightarrow 0$$

for some fixed $C < \infty$. Proposition 7then gives the interval and its strict rankings. If $M_P \geq M_C$, nonnegative activities imply $\mathcal{F}_C^S(0) \subseteq \mathcal{F}_P^S(0)$ and $V_P \geq V_C$, identifying the change as a productive improvement. Finally, both thresholds tend to zero, so every fixed positive fee eventually exceeds both. *Q.E.D.*

APPENDIX F: STANDING BALANCES, RESERVE BORROWING, AND INSTITUTIONAL CHOICE

F.1. *Uniform certificates and primitive verification*

Retain the finite positive-linear economy, complete bilateral access, positive action caps, balanced compensation, $B \equiv 0$, and blocked start at zero. Let original balances be sr, $0 \leq s \leq \bar{s}$, and rented balances ϵr, with $r \gg 0$. A uniform core and completion certificate consists of a fixed finite list of voluntary bundles $u^1, \ldots, u^k$, their sum d, a nonnegative resource release $p = Md$, scalars $0 < \underline{\alpha} \leq \alpha_s \leq \overline{\alpha}$, and at most ℓ voluntary completion bundles $v_s^1, \ldots, v_s^{\ell_s}$. They must sum to a planner optimum $z_s^* = \alpha_s d + \sum_j v_s^j$ and satisfy

$$sr + \alpha_s p + M\sum_{j=1}^{t} v_s^j \geq 0 \quad (0 \leq t \leq \ell_s).$$

The productive case additionally requires $p \neq 0$ and $M\sum_{j=1}^{t} u^j \geq -\kappa p$ for every prefix, for a fixed $\kappa > 0$. The completion may consume original balances and preserves all rented capacity.

The certificate has a finite verification on any interval where its activity bundles are affine. To see this, write the planner objective as $w^{\mathsf{T}}z$. The planner linear program and a dual are

$$\max\{w^{\mathsf{T}}z : Mz + sr \geq 0,\ 0 \leq z \leq \bar{z}\}, \qquad \min\{s\lambda^{\mathsf{T}}r + \mu^{\mathsf{T}}\bar{z} : \mu - M^{\mathsf{T}}\lambda \geq w,\ \lambda, \mu \geq 0\}.$$

Suppose α_s and a fixed list $v_s^1, \ldots, v_s^\ell$ are affine, with zero bundles allowed. Fix their transaction types. Voluntariness, activity caps, nonnegativity, and every completion inequality are linear in s, so checking $s = 0$ and $s = \bar{s}$ checks the entire interval. If one dual-feasible

pair (λ, μ) has objective equal to $W(z_s^*)$ at both endpoints, equality holds throughout the interval. Thus the selected allocation is planner-optimal for every s. The test constructs the certificate from resource effects, preferences, caps, and linear inequalities. A stable planner basis often supplies the affine candidate; the certificate then verifies that its comparative static can be executed.

F.2. *Proof of Theorem 9*

Put $t = s + \epsilon$. The case $t = 0$ has no positive feasible first approval, and the corresponding envelopes are infinite at $\epsilon = 0$. All statements below concern $t > 0$. Planner values satisfy $0 < V(0) \le V(s) \le \bar{V} < \infty$ by monotonicity and compact action caps.

Lower bounds uniform across allocations. Use all account coordinates and choose strictly positive weights λ. A blocked start at zero implies

$$\delta = \min_{d' \in \mathcal{D}} \sum_i \lambda_i [-(Md')_i]_+ > 0, \quad S = \lambda^{\mathsf{T}} r > 0, \quad g = \max\{0, \max_{d' \in \mathcal{D}} \lambda^{\mathsf{T}} M d'\}.$$

These constants concern the technology at zero balances and remain fixed when original capacity is introduced. Write each approval as $x_j d_j$, with $d_j \in \mathcal{D}$, and let H_j be its available weighted stock. Feasibility gives

$$\delta x_j \le H_{j-1}, \qquad H_j \le H_{j-1} + g x_j, \qquad H_0 = tS.$$

Thus the argument of Theorem A.1 applies with initial stock t, irrespective of how much must eventually be repaid. In the nonexpanding case take its certificate as λ, so $g = 0$. Any implemented allocation of surplus $W > 0$ then requires

$$T \ge \frac{\delta W}{\bar{v} S t}.$$

In the productive case $g > 0$: otherwise the positive core release would contradict $\lambda \gg 0$. The analogous bound is

$$T \ge \frac{1}{\log(1 + g/\delta)} \log\left(1 + \frac{gW}{\bar{v} S t}\right).$$

An allocation with $W < V(0)/2$ loses at least $V(0)/2$ in gross surplus relative to its own planner. For every other allocation, the bounds yield constants $a_C, a_P, b_P > 0$, independent of s and ϵ, such that

$$T \geq a_C/t \quad \text{or} \quad T \geq a_P \log(1 + b_P/t),$$

respectively. The elementary inequality $\log(1+bx) \geq \min\{1, b\} \log(1+x)$, $b, x > 0$, makes the productive bound comparable to $\log(1 + 1/t)$. Consequently,

$$L_C(s; \rho, \tau) \geq \min\{V(0)/2, \min(1, a_C)\Phi_C(s; \rho, \tau)\},$$

with the analogous inequality for L_P and Φ_P. Both envelopes converge to zero uniformly in s: their maximum over $s \geq 0$ occurs at $s = 0$. The fixed first term therefore ceases to bind for sufficiently small τ, establishing uniform lower bounds. This step is what permits the institution to choose an arbitrary allocation.

Planner implementation under conservation. Let $A_i = \sum_{j=1}^{k} [-(Mu^j)_i]_+$ and $A = \overline{\alpha} \max_i A_i/r_i$. Choose $n = \max\{1, \lceil A/t \rceil\}$ and repeat the ordered core list n times, scaling every bundle by α_s/n. The temporary shortfall in each repetition is bounded by $\alpha_s A_i/n \leq t r_i$. Completed repetitions release nonnegative capacity. Thus each prefix is feasible and the completed core leaves $tr + \alpha_s p$. Execute the completion list. Its prefix balances equal

$$\epsilon r + \left(sr + \alpha_s p + M \sum_{j=1}^{t} v_s^j \right) \geq \epsilon r.$$

The entire rented reserve is therefore returned. The approval count is at most $k \max\{1, \lceil A/t \rceil\} + \ell$. All partial activity vectors are bounded by the terminal planner vector, so the action caps hold throughout.

Planner implementation with productivity. Put $\sigma = \min_{i:p_i>0} r_i/p_i > 0$ and $\Lambda = 1 + 1/\kappa > 1$. Since $tr \geq \sigma t p$, the geometric construction in Theorem 4implements the core $\alpha_s d$ with at most

$$k \left\lceil \frac{\log(1 + \overline{\alpha}/(\sigma t))}{\log \Lambda} \right\rceil$$

approvals. Truncating the last repetition gives the exact core quantity. The same completion inequalities then return the rented reserve. For any fixed upper bound $t \leq t_0 < \infty$, these two

constructions give constants A_C, A_P such that their respective counts are at most A_C/t and $A_P \log(1+1/t)$; additive constants are absorbed uniformly on this bounded interval.

The minimizers of the displayed envelopes satisfy $t_C = \max\{s, \sqrt{\tau/\rho}\}$ and $t_P = \max\{s, (\sqrt{1+4\tau/\rho}-1)/2\}$. They are bounded by $\bar{s}+1$ for small τ. Evaluating the constructions at these minimizers gives $L_C \le \max\{1, A_C\}\Phi_C$ and $L_P \le \max\{1, A_P\}\Phi_P$. If the original definition requires $\epsilon > 0$, take an arbitrarily small positive reserve when the minimizer is zero; the existing finite schedule is feasible with that extra unused capacity, and the sequential supremum has the same bound.

Evaluation of the envelopes. For conservation the first-order condition in total stock is $\rho = \tau/t^2$. The constraint $t \ge s$ gives the formula in the main text and $\Phi_C \asymp \tau/(s+\sqrt{\tau/\rho})$. For productivity it is $\rho = \tau/[t(1+t)]$. Thus $t_P \sim \tau/\rho$ in the interior, while at $t = s$ the envelope equals $\tau \log(1+1/s)$. These formulas imply, uniformly on $[0, \bar{s}]$,

$$\Phi_P \asymp \tau \log\left(1 + \frac{1}{s+\tau/\rho}\right).$$

This proves the theorem.

F.3. *Borrowing and institutional choice*

Consider $\eta = \tau/\rho \downarrow 0$ and $\eta \ll s \ll \sqrt{\eta}$. The productive construction at $\epsilon = 0$ has loss $O(\tau \log(1/s))$, which matches the optimal order. In a conserving economy, any schedule whose total loss is $O(\rho\sqrt{\eta})$ must have surplus at least $V(0)/2$ for small η. Its lower bound $T \ge a_C/(s+\epsilon)$ then forces $s + \epsilon = \Omega(\sqrt{\eta})$. The reserve cost simultaneously forces $\epsilon = O(\sqrt{\eta})$. Because $s = o(\sqrt{\eta})$, every such schedule has $\epsilon = \Theta(\sqrt{\eta})$. This is a statement about actual implementations attaining the optimal order, rather than an identification of the exact reserve optimizer from an envelope.

For Corollary 10, write $a = \sqrt{\tau/\rho}$. Uniform bounds imply

$$\frac{L_P}{L_C} \le C(s+a)\log\left(1 + \frac{1}{s+a^2}\right).$$

If $s \le a$, the right-hand side is bounded by $2Ca\log(1+a^{-2}) \to 0$. If $s > a$ and $s \to 0$, it is bounded by $2Cs\log(1+s^{-1}) \to 0$. This proves the ratio limit along every path $s(\tau) \to 0$.

For fixed $s > 0$, both envelope minima occur at $\epsilon = 0$ for sufficiently small τ. Uniform constants therefore give

$$L_C \geq c_C \tau / s, \qquad L_P \leq C_P \tau \log(1 + 1/s).$$

Since $s \log(1 + 1/s) \to 0$, for all sufficiently small fixed s the interval

$$(\max\{0, C_P \log(1 + 1/s) - 1\},\ c_C/s - 1)$$

is nonempty. Any f_s in this interval, and the common fee $F = f_s \tau$, satisfy $L_P < \tau + F < L_C$. As $V(s) \geq V(0) > 0$, both joint options and the productive sequential option are adopted for small τ. The comparison follows from the exact joint value $V(s) - \tau - F$. The fee is common across technologies; the planner values need not coincide. For a fixed positive fee independent of τ, both technologies eventually prefer sequential settlement, because both implementation losses tend to zero. The corollary identifies the relevant fee interval at each approval price and its coefficient for fixed standing capacity.

F.4. *Membership of M_h and the true planner allocation*

Fix $0 \leq h < 1/2$ and $0 \leq s \leq 1/32$, and put $D_h = 2 + 3h$. Original capacities are $s\mathbf{1}$. Weighting the pension and payroll inequalities gives

$$b_1 + b_2 \geq \frac{2(1+h)}{D_h} a - \frac{4 - 2h}{D_h} s.$$

For example, multiply the pension inequality by $2/D_h$ and the payroll inequality by $2(1 - h)/D_h$; the supplier coefficients are one in both columns, while the demander coefficient is $-2(1 + h)/D_h$. The resulting employee-surplus bound is

$$W(z) \leq \left(1 - \frac{1+h}{5D_h}\right) a + \frac{2-h}{5D_h} s.$$

Its coefficient on a is positive. The stated $z^*_{h,s}$ attains equality at $a = 1$, respects all caps, and has balances

$$s\mathbf{1} + M_h z^*_{h,s} = \left(0, 0, \frac{h + (6+h)s}{D_h}\right).$$

This proves planner optimality and the value formula. Positive dual weights on pension and payroll also imply uniqueness.

Take $d = (1, 1/2, 1/2)$, whose release is $p = (h/2, 0, 0)$, and $\alpha_s = \theta_s = 2(1 - 8s)/D_h$. The single completion bundle uses supplier 2 and has increments

$$x_s = \frac{3h + 16s}{D_h}, \qquad y_s = \frac{2h + (12 + 2h)s}{D_h}.$$

It is voluntary because

$$x_s - y_s/10 = \frac{(14/5)h + (74/5 - h/5)s}{D_h} > 0$$

whenever $(h, s) \neq (0, 0)$; at $(0, 0)$ the zero completion is deleted. Paying $y_s/10$ from the demander to supplier 2 balances compensation and leaves the supplier indifferent. The proportional construction already supplies voluntary, balanced compensation at every preceding approval.

The accounts after the proportional stage, before the completion, are

$$(s + \epsilon + h\theta_s/2,\ s + \epsilon,\ s + \epsilon).$$

The completion consumes $s + h\theta_s/2$ pension units and s payroll units, and releases $(h + (4 - 2h)s)/D_h$ position units. It therefore leaves

$$\left(\epsilon, \epsilon, \epsilon + \frac{h + (6 + h)s}{D_h}\right).$$

Every account is nonnegative, the reserve is fully repaid, and all caps hold because adjustments are irreversible and their terminal vector is $z^*_{h,s}$. Both endpoints are nonnegative, so linear interpolation also preserves feasibility throughout the atomic approval.

For $h = 0$, any fixed voluntary decomposition of d supplies the divided core construction. For $h > 0$, the core construction in Appendix C supplies the productive prefix inequality. Its constants depend on h and not on s. The scalar θ_s is positive and uniformly bounded on $[0, 1/32]$, and the completion is affine in s. Thus both parts of Theorem 9apply to the true planner optima throughout this interval.

F.5. *Exact conserving frontier and institutional cutoffs*

PROOF OF PROPOSITION 13: Write $t = s + \epsilon$ and $q = W - H$ for the difference between payroll and position balances. Conservation gives $|q| \leq 3t$ at every intermediate endpoint.

Initially $q = 0$; the planner endpoint has $q = -3s$. A supplier-1 increment of size y raises q by $y/2$, and a supplier-2 increment lowers it by $y/2$. Hence a schedule with T approvals satisfies

$$b_1 + b_2 = 2\sum_{j=1}^{T} |q_j - q_{j-1}| \le 12t(T-1) + 6s.$$

This remains a valid upper bound if demander-only transactions occur, as they leave q unchanged. At the planner optimum, $b_1 + b_2 = 1 - 2s$, proving necessity of the stated reserve.

For attainment put

$$r_T = \frac{1-8s}{12(T-1)}, \qquad e = [r_T - s]_+, \qquad t = s + e = \max\{s, r_T\}.$$

Choose an alternating supplier order ending with supplier 2. Set $q_0 = 0$; at every prefinal step set $q_j = 3r_T$ after supplier 1 and $q_j = -3r_T$ after supplier 2; finally set $q_T = -3s$. Choose supplier increments $y_j = 2|q_j - q_{j-1}|$. Choose the associated demander increment to make the pension balance zero at every prefinal endpoint and equal to e at the last endpoint. At each prefinal endpoint, payroll and position balances are $(3t + q_j)/2$ and $(3t - q_j)/2$, respectively. These are nonnegative since $r_T \le t$. At completion the balances are $(e, e, e + 3s)$.

The supplier quantities sum to $12r_T(T-1) + 6s = 1 - 2s$ and their difference is $-6s$. Thus their terminal quantities are exactly $(1/2 - 4s, 1/2 + 2s)$. The pension identity then gives demander quantity one. Interior demander increments are $3y_j/4$ for supplier 1 and $5y_j/4$ for supplier 2. The first increment adds $t/2$ to the respective expression. The last increment is $5y_T/4 - e/2$. If $r_T \ge s$, the latter equals $7r_T + 8s$ and $y_T = 6(r_T + s)$; if $r_T < s$, it equals $5y_T/4$. All are strictly larger than $y_j/10$. Paying each supplier $y_j/10$ from its demander therefore balances compensation and satisfies participation. Every activity increment is nonnegative; monotonicity and the terminal planner vector imply all action caps. The final balances return the full rented reserve. This proves attainment. Solving $\epsilon_T^*(s) = 0$ gives the stated no-rental approval count. *Q.E.D.*

Optimization over allocations. For arbitrary $\rho, \tau > 0$, the optimized conserving value is exactly

$$\max\left\{0, \frac{37s}{5} - \tau, \sup_{n\geq 1}\left[\frac{9}{10}a_n + \frac{s}{5} - \rho\left[\frac{a_n - 8s}{12n} - s\right]_+ - \tau(n+1)\right]\right\},$$

where

$$a_n = \begin{cases} 1, & \rho \leq 54n/5, \\ \min\{1, (12n+8)s\}, & \rho > 54n/5. \end{cases}$$

A one-approval allocation uses at most one supplier. Supplier 2 attains the maximum at $(a, b_2) = (8s, 6s)$ without rented capacity, giving the second term in the maximum. A reserve cannot enlarge the attainable terminal set for one approval because it must be repaid at that same endpoint.

For $T = n + 1 \geq 2$, the planner dual gives $W \leq 9a/10 + s/5$, and total variation gives rented reserve at least $[(a - 8s)/(12n) - s]_+$. The slope of the resulting net-value bound as a function of a is $9/10$ below its zero-rental kink and $9/10 - \rho/(12n)$ above it. Its maximizer is a_n. Since $a_n \geq 8s$, the frontier construction just proved, with 1 replaced by a_n, attains this bound. For $a < 8s$, the dual gross-surplus bound is smaller than the feasible one-approval value before costs; such allocations cannot improve the maximum. At $s = 0$ any zero-quantity term is discarded in favor of the baseline. This proves the formula over all allocations and schedules.

PROOF OF COROLLARY 14: At $\rho = 12$, the $n = 1$ optimum is $a_1 = 20s$, whereas every $n \geq 2$ chooses $a_n = 1$. The one-approval and two-approval net values are $37s/5 - \tau$ and $91s/5 - 2\tau$, respectively. Both are strictly below $9/10 + s/5 - 101/10000$ throughout $s \in [0, 1/32]$. The baseline is also below that positive joint value. For $n \geq 2$, it remains to compare

$$C_n(s) = \left[\frac{1 - 8s}{n} - 12s\right]_+ + \frac{n+1}{10000}$$

with the joint cost $101/10000$.

At $s_J = 1/1208$, the $n = 100$ cost equals $101/10000$ with zero rental. Put $A_J = 1 - 8s_J = 150/151$. For every $n \le 99$ rental remains positive and

$$C_n(s_J) - \frac{101}{10000} = (100 - n)\left(\frac{A_J}{100n} - \frac{1}{10000}\right) > 0.$$

For $n \ge 101$, approval costs alone exceed the joint cost. Since all C_n are nonincreasing, joint settlement strictly dominates for $s < s_J$.

The $n = 99$ cost first equals $101/10000$ at $s_S = 9901/11960000$. For every $n \le 98$,

$$C_n(s_S) - C_{99}(s_S) = (99 - n)\left(\frac{1 - 8s_S}{99n} - \frac{1}{10000}\right) > 0.$$

The $n = 100$ cost remains exactly $101/10000$ for $s \ge s_J$, whereas larger counts cost more in approvals alone. This proves the precise tie interval. For $s > s_S$, $C_{99}(s) < 101/10000$, proving strict sequential dominance.

In $M_{1/4}$ the true planner construction uses proportional quantity $\theta_s = 8(1 - 8s)/11 \le 8/11$. Scale the existing 16-approval proportional construction by $1 - 8s$ and use its seed from original balances first, renting any remaining shortfall. This rents at most $1/144309$. The supplier-2 completion reaches $z^*_{1/4,s}$ in at most 17 approvals. Its total implementation cost is bounded by

$$\frac{17}{10000} + \frac{12}{144309} < \frac{101}{10000}.$$

The productive sequential value therefore strictly exceeds the joint value at every $s \in [0, 1/32]$. ***Q.E.D.***

F.6. *Exact verification of the fixed positive-balance comparison*

Set $s = 1/2500$, $\rho = 12$, $\tau = 1/10000$, and $F = 1/100$. For $h = 0$,

$$V^*_{0,s} = \frac{11251}{12500}, \qquad \mathcal{V}^{\text{joint}}_{0,s} = \frac{44499}{50000}.$$

The preceding global formula selects $n = 100$. Indeed, the untruncated objective has consecutive differences $\tau - (1 - 8s)/[n(n + 1)]$, which change sign between $n = 99$ and $n = 100$;

its reserve remains positive at the minimizer. Zero-rental counts start at $n = 208$ and cost at least $209/10000$. The one- and two-approval values are smaller. Consequently

$$\epsilon^* = \frac{323}{750000}, \qquad T^* = 101,$$
$$\mathcal{V}_{0,s}^{\text{seq}} = \frac{11251}{12500} - 12\frac{323}{750000} - \frac{101}{10000} = \frac{221203}{250000},$$
$$0.884812 < \frac{44499}{50000} = 0.889980.$$

For $h = 1/4$,

$$z_{1/4,s}^* = \left(1, \frac{2492}{6875}, \frac{7509}{13750}\right), \quad V_{1/4,s}^* = \frac{125007}{137500}, \quad \mathcal{V}_{1/4,s}^{\text{joint}} = \frac{494473}{550000}.$$

The proportional target has $\theta_s = 4984/6875$. At $T = 9$, the exact frontier gives

$$\epsilon_9(1/4, \theta_s) = \frac{(1/4)(4984/6875)}{6(3^4 - 1)} = \frac{623}{1650000} < \frac{1}{2500} = s.$$

Use that nine-approval construction entirely from original capacity, leaving the difference $s - 623/1650000$ unused in each account. The completion has

$$(x_s, y_s) = \left(\frac{1891}{6875}, \frac{101}{550}\right), \qquad \text{compensation} = \frac{101}{5500}.$$

Its demander surplus is positive and its supplier surplus is zero. No reserve is rented, and the ten approvals give

$$\mathcal{V}_{1/4,s}^{\text{seq}} \geq \frac{125007}{137500} - \frac{10}{10000} = \frac{249739}{275000} = 0.9081418181\ldots > \frac{494473}{550000} = 0.8990418181\ldots.$$

The construction reaches the new planner optimum exactly. The conserving value is optimized over all admissible allocations, reserves, and schedules, so these inequalities prove the reversal at fixed positive original balances and fixed prices.

APPENDIX G: EXACT VERIFICATION OF THE ENDOGENOUS-ALLOCATION REVERSAL

This section verifies the productive witness in Theorem 12using exact arithmetic. Put $h = 1/4$, $\theta = 8/11$, and $e = 1/144309$. The exact frontier gives

$$\gamma_{1/4} = 3, \qquad G_{16}(1/4) = 2 \cdot 3^7 = 4374, \qquad \epsilon_{16}(1/4, 8/11) = \frac{(1/4)(8/11)}{6(4374 - 1)} = e.$$

The reserve is e in each account. As throughout the paper, the price ρ applies to the scalar reserve intensity e, not separately to each coordinate; at $\rho = 12$ its cost is $12e$.

G.1. *The first sixteen approvals*

Write (x_t, y_t) for the demander and participating supplier increments. Starting from zero activity and balances (e, e, e), use the following list:

t	supplier	x_t/e	y_t/e
1	1	11	12
$2m, \quad m = 1, \dots, 7$	2	$30 \cdot 3^{m-1}$	$24 \cdot 3^{m-1}$
$2m+1, \quad m = 1, \dots, 7$	1	$42 \cdot 3^{m-1}$	$48 \cdot 3^{m-1}$
16	2	26245	26244

The rows indexed by m are interleaved in increasing approval order. For supplier 1 a pair has resource effect $(-2x + 7y/4, x - y/2, x - y)$; for supplier 2 it has effect $(-2x + 5y/2, x - 3y/2, x - y)$. Substitution gives the successive balances

$$\begin{aligned} q^1 &= (0, 6e, 0), \\ q^{2m} &= (0, 0, 6 \cdot 3^{m-1} e), && m = 1, \dots, 7, \\ q^{2m+1} &= (0, 18 \cdot 3^{m-1} e, 0), && m = 1, \dots, 7, \\ q^{16} &= (13120e, e, e) = (e + 1/11, e, e). \end{aligned}$$

Every coordinate is nonnegative. The sum of the first sixteen increments is

$$z^{16} = (104952e, 52476e, 52476e) = \left(\frac{8}{11}, \frac{4}{11}, \frac{4}{11} \right).$$

All increments are nonnegative, so no earlier activity exceeds these totals; the unit caps therefore hold. These calculations also verify directly that this schedule realizes the exact-frontier first stage.

G.2. *Completion at the planner optimum and repayment*

Approval 17 uses supplier 2 and increments

$$(x_{17}, y_{17}) = (3/11, 2/11), \qquad M_{1/4}(3/11, 0, 2/11)^{\mathsf{T}} = (-1/11, 0, 1/11)^{\mathsf{T}}.$$

It consumes exactly the pension capacity released by the proportional target. The final activity and balances are

$$z^{17} = \left(1, \frac{4}{11}, \frac{6}{11}\right) = z^*_{1/4}, \qquad q^{17} = (e, e, e + 1/11).$$

The unit caps hold, and repayment of e in each account leaves $(0, 0, 1/11)$, precisely the planner's residual ledger. Linear resource effects make every balance within an approval a convex combination of its endpoint balances, so intermediate feasibility holds even if an approval is interpreted as continuous implementation.

At every approval, pay the participating supplier $y_t/10$ and charge the demander the same amount. The compensation vector sums to zero. The supplier receives its exact incremental cost, while the demander's gain is $x_t - y_t/10$. For the four rows of the first-stage list these gains, divided by e, are respectively

$$49/5, \qquad (138/5)3^{m-1}, \qquad (186/5)3^{m-1}, \qquad 118103/5.$$

All are strictly positive. Approval 17 gives the demander $14/55 > 0$. Thus every step is voluntary with enforceable balanced compensation; no public subsidy or decentralized bargaining process is used.

G.3. *Merging the final approvals*

Approvals 16 and 17 involve the same pair. Replacing them by their summed increment preserves participation, balanced compensation, the terminal allocation, and nonnegative intermediate balances. It therefore gives a 16-approval witness at the same reserve, with net value

$$\frac{436442249}{481030000} + \frac{1}{10000} = \frac{27280647}{30064375} \simeq 0.907408.$$

The main table uses this stronger 16-approval bound. The preceding 17-stage list separates the proportional-target construction from planner completion. Rounded decimals are descriptive; all inequalities use exact fractions.

APPENDIX H: COMMITMENT COSTS THAT DEPEND ON THE ALLOCATION

The authorization technology in the main text permits certification coverage to vary with the allocation. The resulting charge $F(z)$ prices the additional resources used by that joint program, at common administrative prices across technologies.

A constant fee is not essential to the reversal. Let $F(z)$ be any common nonnegative fee rule for nonzero allocations, used without change across the two technologies. Write

$$\mathcal{V}_h^J[F] = \max\left\{0, \sup_{z \in \mathcal{F}_h^S(0)\setminus\{0\}} [W(z) - \tau - F(z)]\right\}, \qquad \mathcal{F}_h^S(0) = \{z \in X : M_h z \geq 0\}.$$

The supremum accommodates fee rules for which a maximizing allocation need not exist. Sequential compensation and implementation prices are unchanged.

PROPOSITION A.5—Reversal with allocation-dependent commitment costs: *In the two economies and at the reserve and approval prices of Theorem 12, suppose every nonzero feasible allocation has a fee satisfying*

$$\frac{152309}{96206000} < \underline{F} \leq F(z) \leq \overline{F} < \frac{1}{50}.$$

Then

$$\mathcal{V}_0^J[F] > \mathcal{V}_0^{\text{seq}}, \qquad \mathcal{V}_{1/4}^{\text{seq}} > \mathcal{V}_{1/4}^J[F].$$

In particular, any common fee rule taking values in $[0.005, 0.015]$ *preserves the strict reversal. The joint institution need not choose a frictionless planner optimum.*

PROOF: In the conserving economy, implementing its planner optimum jointly gives at least $9/10 - \tau - \overline{F} > 8799/10000$, the globally optimized sequential value. In the productive economy every joint allocation gives at most $10/11 - \tau - \underline{F}$. The merged 16-approval schedule gives

$$\mathcal{V}_{1/4}^{\text{seq}} \geq \frac{10}{11} - \overline{C}_P, \qquad \overline{C}_P = \frac{12}{144309} + \frac{16}{10000} = \frac{50603}{30064375}.$$

The lower fee bound is precisely $\underline{F} > \overline{C}_P - \tau$. Thus this positive sequential value exceeds both the best nonzero joint value and the zero-cost baseline. *Q.E.D.*

For an example in which the joint allocation changes, let

$$F(z) = \frac{1}{200} + \left[a - \frac{99}{100}\right]_+, \qquad [x]_+ = \max\{x, 0\}, \quad z \neq 0.$$

This continuous nondecreasing fee belongs to the stated interval. For each a, the joint institution uses the cost-minimizing supplier mix in Proposition 1. Its objective is $V_h^* a - \tau - F(z)$, increasing up to $a = 99/100$ and decreasing thereafter, since $0 < V_h^* < 1$. Consequently its unique allocation is $(99/100)z_h^*$ in each economy. Joint net values are 0.885900 at $h = 0$ and 0.894900 at $h = 1/4$, and the reversal persists. The administrative coverage charge changes the selected quantity, while a common resource-price schedule preserves the ranking across technologies.

APPENDIX I: ALLOCATION DISTORTION AND BINDING PARTICIPATION

Settlement costs can change the allocation itself, and employee participation can tighten the physical reserve frontier. The following examples separate these economic margins. They retain M_0, the $B \equiv 0$ mandate, complete information, divisibility, irreversibility, and employee-balanced compensation. Only preferences and implementation prices are varied explicitly.

PROPOSITION A.6—Allocation distortion and a participation frontier: *Let initial balances be zero, let all action caps be one, and require full reserve repayment in the conserving economy M_0.*

(i) Suppose

$$U(a) = a - \frac{2}{5}a^2, \qquad C_j(b_j) = \frac{b_j}{10}, \qquad \rho = \frac{36}{5}, \quad \tau = \frac{1}{20}, \quad F = \frac{1}{100}, \quad \mathcal{K} = 0.$$

The gross planner optimum and the joint-settlement optimum are $z_1 = (1, 1/2, 1/2)$. The unique sequentially optimal allocation is instead $z_{7/8} = (7/8, 7/16, 7/16)$, implemented with four approvals and reserve $7/288$ in each account. Optimized net surplus is $11/25$ under joint settlement and $17/160$ under sequential settlement.

(ii) Suppose instead that $U(a) = a$ and $C_j(b_j) = cb_j$, where $0 < c < 1$. For the target z_1, the least reserve permitting voluntary implementation with at most two approvals is

$$\epsilon_2^{\text{vol}}(c) = \max\left\{\frac{1}{12}, c - \frac{3}{4}\right\}.$$

The physical lower bound $1/12$ is attainable voluntarily if and only if $c \leq 5/6$. For $c > 5/6$, participation binds and raises the reserve requirement despite positive gross target surplus $1 - c$.

PROOF OF PROPOSITION A.6: Since every column of M_0 sums to zero, terminal feasibility after repayment forces $M_0 z = 0$. Thus all feasible allocations are $z_\theta = (\theta, \theta/2, \theta/2)$, $0 \leq \theta \leq 1$.

Part (i). Gross surplus is $W(z_\theta) = 9\theta/10 - 2\theta^2/5$, with derivative $9/10 - 4\theta/5 \geq 1/10$. The gross planner therefore uniquely chooses $\theta = 1$. Joint settlement charges the constant $\tau + F = 3/50$ for a nonzero allocation, so it also chooses $\theta = 1$, yielding $1/2 - 3/50 = 11/25 > 0$. The physical frontier in Proposition 2

requires $e \geq \theta/[12(T-1)]$. Its attaining schedule has ratios $x/y = 5/6$ initially, $5/4$ or $3/4$ internally, and $2/3$ or $7/6$ finally. Hence $x/y \geq 2/3$. Because $U'(a) = 1 - 4a/5 \geq 1/5$, every approval satisfies $\Delta U \geq x/5 \geq 2y/15 > y/10 = \Delta C$. Paying the supplier $y/10$ from the demander verifies balanced compensation and participation, so the physical frontier remains exact.

For $n = T - 1 \geq 1$, net surplus at that frontier is $f_n(\theta) = (9/10 - 3/(5n))\theta - 2\theta^2/5 - (n+1)/20$. Its unique positive maximizer is $\theta_n = \min\{1, 9/8 - 3/(4n)\}$. The optimized value, extended to real $n \geq 1$, is

$$H(n) = \begin{cases} \dfrac{5}{8}\left(\dfrac{9}{10} - \dfrac{3}{5n}\right)^2 - \dfrac{n+1}{20}, & 1 \leq n \leq 6, \\ \dfrac{1}{2} - \dfrac{3}{5n} - \dfrac{n+1}{20}, & n \geq 6. \end{cases}$$

The branches and first derivatives agree at 6. Moreover, $H''(n) = 27(1-n)/(20n^4) \leq 0$ for $1 < n < 6$, and $H''(n) = -6/(5n^3) < 0$ for $n > 6$. Thus H is concave, strictly so above 1. Its values $H(2) = 3/40$, $H(3) = 17/160$, and $H(4) = 13/128$ satisfy $H(3) - H(2) = 1/32 > 0$ and $H(4) - H(3) = -3/640 < 0$. Nonincreasing discrete differences establish the unique integer optimum $n = 3$; its positive value exceeds the baseline. Consequently $\theta = 7/8$, $T = 4$,

and $e = (7/8)/(12 \cdot 3) = 7/288$. Gross surplus $77/160$ less reserve cost $7/40$ and approval cost $1/5$ gives $17/160$. The attaining schedule preserves caps, irreversibility, intermediate feasibility, and repayment; compensation was verified above.

Part (ii). Two approvals must each use one of the suppliers, with supplier increment $1/2$; there is no spare demander-only approval. If x is the first demander increment, balanced compensation at both approvals requires $x \geq c/2$ and $1 - x \geq c/2$. The first account vector is

$$\begin{cases} (e - 2x + 3/4,\ e + x - 1/4,\ e + x - 1/2), & \text{supplier 1 first,} \\ (e - 2x + 5/4,\ e + x - 3/4,\ e + x - 1/2), & \text{supplier 2 first.} \end{cases}$$

In the first order, pension feasibility gives $e \geq 2x - 3/4 \geq c - 3/4$. In the second, payroll feasibility gives $e \geq 3/4 - x \geq c/2 - 1/4 \geq c - 3/4$, using $c \leq 1$. The physical bound also requires $e \geq 1/12$. Every voluntary schedule therefore needs $e \geq \max\{1/12, c - 3/4\}$.

To attain this bound, use supplier 1 first with $x = \max\{5/12, c/2\}$ and $e = \max\{1/12, c - 3/4\}$. For $c \leq 5/6$, the first balances are $(0, 1/4, 0)$ and the second demander increment is $7/12$; both participation inequalities hold. For $c > 5/6$, the first balances are $(0, 3c/2 - 1, 3c/2 - 5/4) \geq 0$, and the second demander increment is $1 - c/2 \geq c/2$. Pay each supplier $c/2$ from the demander. In this latter case the first demander gain is zero and the second is $1 - c > 0$. Both increments respect the caps; linear account effects preserve feasibility between endpoints. Since $M_0 z_1 = 0$, final balances are (e, e, e), permitting full repayment.

At $c = 9/10$, the attaining reserve is $3/20$ and the transaction vectors are $(9/20, 1/2, 0)$ and $(11/20, 0, 1/2)$. Balances move through $(0, 7/20, 1/10)$ to $(3/20, 3/20, 3/20)$. Thus $1/12$ is insufficient despite positive target surplus $1/10$. ***Q.E.D.***

APPENDIX J: A FIXED-GRID REVERSAL WITH POSITIVE ORIGINAL BALANCES

PROOF OF COROLLARY 15: *Productive grid implementation.* Set $\delta = 1/500$ and $e = \delta/4$. Starting from $e\mathbf{1}$, execute eight rounds with integer quantities $(u_1, \ldots, u_8) = (1, 2, 4, 8, 16, 32, 64, 56)$. Each round uses supplier 1 and then supplier 2, with demander and supplier increments $(u_j\delta, u_j\delta)$ at each approval. From balances (p, e, e), the successive endpoints are

$$(p - u_j\delta/4, e + u_j\delta/2, e), \qquad (p + u_j\delta/4, e, e).$$

Thus a round is feasible if $u_j\delta \le 4p$. Initially $p = \delta/4$; full rounds double p, and the last round's 56 units are below the available 128. Since $\sum_j u_j = 183$, the sixteen approvals reach $\widehat{z}_\delta = (366/500, 183/500, 183/500)$ and balances $(e + 183/2000, e, e)$. A final supplier-2 approval has $(\Delta a, \Delta b_2) = (134/500, 89/500)$ and resource effects $(-91/1000, 1/1000, 9/100)$. Hence

$$z_\delta = (1, 183/500, 272/500), \quad M_{1/4} z_\delta = (1/2000, 1/1000, 9/100) \ge 0.$$

All prefixes respect the caps because activities increase to $z_\delta \le \mathbf{1}$; linear interpolation preserves account feasibility. Paying each supplier its incremental cost and charging the demander the same amount gives exact balance. Demander gains are $9u_j\delta/10$ in the rounds and $1251/5000$ at the final approval, so all trades are voluntary. Terminal balances are $e\mathbf{1} + M_{1/4} z_\delta$, permitting full repayment. Adding any $R^0 \ge 0$ preserves every constraint. Gross surplus is $909/1000$, and $12e + 17/10000 = 77/10000$, giving net value $9013/10000$.

Global conserving bound. Relax the grid and start from original balances $s\mathbf{1}$, with rented reserve $\epsilon\mathbf{1}$. For any T-approval schedule in M_0, write a and $b = b_1 + b_2$ for terminal activities and $q_t = W_t - H_t$ for the difference between payroll and position balances. Conservation gives $P_t + W_t + H_t = 3(s+\epsilon)$, so $|q_t| \le 3(s+\epsilon)$ and $q_0 = 0$. Repayment leaves nonnegative balances summing to $3s$, hence $|q_T| \le 3s$. Each supplier increment changes q_t by half its quantity, with sign determined by supplier identity. For $T \ge 2$,

$$b = 2\sum_{t=1}^{T} |q_t - q_{t-1}| \le 12(s+\epsilon)(T-1) + 6s.$$

The terminal position balance gives $a \le b + 2s$ and $W(z) \le 9a/10 + s/5$. Putting $n = T - 1$ therefore yields

$$W(z) - 12\epsilon - \frac{T}{10000} \le a\left(\frac{9}{10} - \frac{1}{n}\right) - \frac{n+1}{10000} + \left(\frac{61}{5} + \frac{8}{n}\right) s.$$

The bound remains valid if the implied lower bound on ϵ is negative. For $n = 1$, maximizing the first two terms over $a \in [0,1]$ gives $-2/10000$. For $n \ge 2$, their maximum is $8799/10000$, since $1/n + (n+1)/10000$ is minimized at $n = 100$ with value $201/10000$. The coefficient

on s is at most $101/5$. For $T = 1$, $b = 2|q_T| \leq 6s$ and $a \leq 8s$, so net value is at most $8s - 1/10000$, below the same bound. Including the baseline,

$$\mathcal{V}_0^{\text{seq}}(s\mathbf{1}) \leq \frac{8799}{10000} + \frac{101}{5}s.$$

This also bounds grid settlement; at $\bar{s} = 1/2500$ its right side is $44399/50000$.

Joint values and unequal original balances. Conserving gross surplus is at most $9/10 + s/5$. At $h = 1/4$, weight the pension and payroll constraints by $8/11$ and $6/11$ to obtain

$$b_1 + b_2 \geq \frac{10}{11}a - \frac{14}{11}s, \qquad W(z) \leq \frac{10}{11} + \frac{7}{55}s.$$

Thus productive joint value, even with divisible activities, is at most $98889/110000 + (7/55)s$. On the grid, gross surplus is a multiple of $1/5000$, and

$$\frac{9}{10} + \frac{\bar{s}}{5} < \frac{9}{10} + \frac{1}{5000}, \qquad \frac{10}{11} + \frac{7}{55}\bar{s} < \frac{909}{1000} + \frac{1}{5000}.$$

The grid planner values are consequently bounded by $9/10$ and $909/1000$. The zero-ledger allocations $(1, 1/2, 1/2)$ and z_δ attain these bounds. Subtracting the common positive-package cost $101/10000$ gives exact joint values $8899/10000$ and $8989/10000$, both above the baseline.

Attainable values are nondecreasing in each original balance: adding a nonnegative vector to all ledgers preserves feasibility, participation, costs, and repayment. For every $R^0 \in [0, \bar{s}]^3$, zero-ledger witnesses remain available and $\bar{s}\mathbf{1}$ supplies the upper bounds. The conserving and productive strict margins are respectively $8899/10000 - 44399/50000 = 6/3125$ and $9013/10000 - 8989/10000 = 3/1250$. This proves the corollary without identifying either optimized sequential value. *Q.E.D.*

APPENDIX K: VALUED PUBLIC BALANCES: FINANCING AND RECOVERY

The main text retains $B \equiv 0$. Here $q = R + Mz$ denotes terminal capacity after reserve repayment, and $B : \mathbb{R}_+^K \to \mathbb{R}$ is concave, continuously differentiable, and coordinatewise nondecreasing; derivatives at the boundary are defined by an extension to a neighborhood. The social benchmark is

$$\mathcal{V}_B(R) = \max_{z \in \mathcal{F}^S(R)} \{W(z) + B(R + Mz) - B(R)\}. \tag{S_B}$$

Compactness gives an optimum z_B^*, and the baseline gives nonnegative value. Implementation expenditures still come from separately authorized operating resources and are deducted from this objective.

PROPOSITION A.7—Financing and participation: *A feasible allocation z can make every employee and the public authority weakly better off precisely when its authorized net public payment s to employees satisfies*

$$-W(z) \leq s \leq B(R + Mz) - B(R). \tag{K.1}$$

Thus unrestricted financing implements a social optimum voluntarily. Employee-balanced compensation requires $W(z) \geq 0$; public participation additionally requires $B(R + Mz) \geq B(R)$.

PROOF: Employee participation requires $W(z) + s \geq 0$ and suffices: for $\alpha_h \geq 0$ summing to one, set $t_h = \alpha_h[W(z) + s] - f_h(z_h)$. Transfers sum to s and employee payoffs are nonnegative. Public participation gives the upper bound. The interval is nonempty whenever social gain is nonnegative. *Q.E.D.*

The interval concerns compensation before implementation costs. If public participation must separately cover expenditure D, subtract D from its upper endpoint. Resource capacity does not itself authorize numeraire payments. For example, take $U(a) = a/2$, $C(b) = b$, unit caps, $R = (0, 1, 1)$, and all fiscal coefficients in (2)equal to one. With $B(q) = 2q_P$, social gain is $b - 3a/2$ on $b \geq a$. Its unique optimum $(0, 1)$ has employee surplus -1 and public gain 2: it requires a subsidy $s \in [1, 2]$ and cannot be employee-balanced.

PROPOSITION A.8—Unrestricted competitive implementation: *Every social optimum has prices $p \geq 0$ at which employee h maximizes $f_h(z_h) + p^\mathsf{T} m_h z_h$ on X_h, the authority maximizes $B(q) - p^\mathsf{T} q$ on $q \geq 0$, and $q = R + Mz$. Conversely, every such equilibrium is socially optimal. Payments*

$$t_h = p^\mathsf{T} m_h z_h, \qquad t_0 = p^\mathsf{T}(R - q) \tag{K.2}$$

balance across employees and the authority and satisfy participation. When $B \equiv 0$, fixed nonnegative employee rebates totaling $p^\mathsf{T} R$ make compensation balance among employees.

PROOF: At an optimum put $q^* = R + Mz^*$ and $\beta = \nabla B(q^*) \geq 0$. The polyhedral normal-cone condition supplies $\lambda \geq 0$ with $\lambda^{\mathsf{T}} q^* = 0$ and

$$\nabla W(z^*) + M^{\mathsf{T}}(\beta + \lambda) \in N_X(z^*).$$

Set $p = \beta + \lambda$. Product structure and concavity give every employee's maximizing condition. For $q \geq 0$, concavity also gives

$$B(q) - p^{\mathsf{T}} q - [B(q^*) - p^{\mathsf{T}} q^*] \leq -\lambda^{\mathsf{T}} q \leq 0.$$

Thus the authority maximizes and markets clear. No strict feasibility is needed for the polyhedral normal-cone representation. Conversely, summing the maximizing inequalities against any feasible allocation cancels all price terms and proves social optimality. Employees can choose zero and the authority can choose $q = R$, proving participation; payments balance by market clearing. If $B \equiv 0$, then $p^{\mathsf{T}} q^* = 0$, so employee payments sum to $-p^{\mathsf{T}} R$. Rebating this nonnegative receipt preserves choices and participation and gives employee balance. *Q.E.D.*

These are supporting-price benchmarks with joint effectiveness, not decentralized bargaining. Adding B does not alter any given target's resource frontier under unchanged employee participation. It can alter the chosen target and invalidate an employee-voluntary decomposition: social optimality gives only $\nabla W(z_B^*)^{\mathsf{T}} z_B^* + \nabla B(R + Mz_B^*)^{\mathsf{T}} Mz_B^* \geq 0$.

PROPOSITION A.9—Recovery with employee financing: *Retain the access and preference assumptions of Theorem A.2. If a social optimum $z_B^* = (a^*, b^*)$ satisfies*

$$\Gamma(z_B^*) = \sum_i U_i'(a_i^*) a_i^* - \sum_j C_j'(b_j^*) b_j^* \geq 0,$$

then every reserve $\epsilon r \gg 0$ permits exact employee-voluntary bilateral implementation and full repayment. A decomposition into $L \leq |D||S| + |D|$ bundles u^ℓ gives an approval bound

$$L \max\{1, \lceil H_B/\epsilon \rceil\}, \qquad H_B = \max_k \frac{\sum_{\ell=1}^{L} [-(Mu^\ell)_k]_+}{r_k}.$$

If public participation is imposed, also require $B(R + Mz_B^*) \geq B(R)$.

PROOF: Put $g_i = U_i'(a_i^*) \geq 0$, $c_j^* = C_j'(b_j^*) > 0$, $Q = \sum_i g_i a_i^*$, and $C = \sum_j c_j^* b_j^*$. If $C > 0$, then $Q \geq C$ and the pair bundles

$$x_{ij} = a_i^* \frac{c_j^* b_j^*}{C}, \qquad y_{ij} = b_j^* \frac{g_i a_i^*}{Q}$$

sum to the target and satisfy $g_i x_{ij} \geq c_j^* y_{ij}$. A zero supplier increment is a demander-only bundle. If $C = 0$, strict positivity of supplier marginal costs gives $b^* = 0$; use demander-only bundles. Omit zero bundles. Execute the list divided by $n = \max\{1, \lceil H_B/\epsilon \rceil\}$ in n rounds. Round-start ledgers lie on the feasible segment from R to $R + Mz_B^*$, plus the reserve. Within-round decreases are at most $\sum_\ell [-(Mu^\ell)_k]_+/n \leq \epsilon r_k$, so balances stay nonnegative and activities never exceed their target quantities. Concavity and convexity make each benefit increment at least $g_i x_{ij}/n$ and its paired cost at most $c_j^* y_{ij}/n$. Reimbursing that cost gives balanced voluntary compensation. The terminal ledger is $R + Mz_B^* + \epsilon r$, permitting repayment. *Q.E.D.*

The condition is sufficient, not necessary. A terminal subsidy satisfying (K.1) alone does not specify financing for intermediate approvals. Such a subsidy requires authorized payment timing and any necessary numeraire bridge finance. Likewise, public participation at every approval adds $\Delta B_t \geq \Delta s_t$; only terminal public participation is considered here.

EXAMPLE A.10—Public value changes the planner allocation: For $M_{1/4}$ and $B_\lambda(q) = \lambda q_P$, $\lambda \geq 0$, the social optimum is $(1, 4/11, 6/11)$ when $\lambda < 4/55$ and $(1, 1/2, 1/2)$ when $\lambda > 4/55$; at equality their joining segment is optimal. The value is $\max\{10/11, 9/10 + \lambda/8\}$. The institutional reversal at the prices of Theorem 12survives for every $\lambda \geq 0$.

PROOF: At fixed $a > 0$, the feasible supplier triangle has vertices $a(4/11, 6/11)$, $a(1/2, 1/2)$, and $a(2/3, 1/3)$, with social values $10a/11$, $a(9/10 + \lambda/8)$, and $9a/10$, respectively. Linearity gives the stated mix, the positive maximal coefficient gives $a = 1$, and equality of the first two coefficients gives $\lambda = 4/55$. For the first optimum, use the merged 16-approval witness of Theorem 12, whose cost is $12/144309 + 16/10000 < 101/10000$. For the second, the exact frontier implements z_1 at cost $9371/5466250 < 101/10000$. Both return the reserve, preserve employee participation, and cost less than the best joint implementation of the same social optimum. In M_0, every repayable terminal balance is zero,

so both optimized institutional values remain unchanged. These comparisons prove the reversal. *Q.E.D.*

APPENDIX L: DISCRETE ADJUSTMENTS AND FINITE APPROVAL BUDGETS

This extension concerns the conserving economy M_0. It retains complete information, irreversible actions, balanced enforceable compensation, $B \equiv 0$, and full reserve repayment. Transfers and reserve capacity remain divisible.

A finite approval budget. With reserve $\epsilon\mathbf{1}$ and at most $N \geq 2$ approvals, repayment restricts terminal allocations to $z_\theta = (\theta, \theta/2, \theta/2)$. The exact frontier in Proposition 2gives

$$\theta_N(\epsilon) = \min\{1, 12\epsilon(N-1)\}, \qquad V_N(\epsilon) = \frac{9}{10}\theta_N(\epsilon).$$

Here V_N is gross employee surplus. Necessity follows from the frontier's lower bound, and its attaining schedule proves sufficiency. The gross welfare loss is $\frac{9}{10}\max\{0, 1 - 12\epsilon(N-1)\}$; the planner optimum requires exactly $1/[12(N-1)]$ reserve per account. One approval cannot implement a positive repayable allocation, which uses both suppliers. Small reserves therefore require a sufficiently large approval budget.

A fixed adjustment unit. Require activity increments in $\delta\mathbb{Z}_+$, $\delta > 0$; an approval may authorize several units. Let $E_N(\theta, \delta)$ be the infimum reserve per account implementing z_θ in at most N grid approvals, and let E_∞ allow any finite number. An infeasible requirement has value $+\infty$. Exact grid implementation requires $\theta/(2\delta)$ to be a positive integer.

PROPOSITION A.11—A discrete reserve floor and the divisible limit: *For a compatible target, $E_\infty(\theta, \delta) = \delta/2$, attained with θ/δ approvals. For each fixed $N \geq 2$,*

$$E_N(\theta, \delta) \geq \max\left\{\frac{\delta}{2}, \frac{\theta}{12(N-1)}\right\}, \qquad E_N(\theta, \delta) \leq \frac{\theta}{12(N-1)} + 4\delta$$

where the upper bound holds for all sufficiently fine compatible grids. Consequently, for $\delta_m = \theta/(2m)$, $E_N(\theta, \delta_m) \to \theta/[12(N-1)]$.

PROOF: Every grid transaction's resource effects are multiples of $\delta/2$. From reserve below $\delta/2$, a feasible first transaction could have no negative effect; conservation would

force every effect to zero. No nonzero bilateral transaction has this property: the position row requires $x = y$, leaving pension effect $-y/2$ or $y/2$. A demander-only transaction also cannot have zero effects. Hence the reserve is at least $\delta/2$. To attain it, alternate supplier-1 and supplier-2 transactions $(x, y) = (\delta, \delta)$. From $(\delta/2, \delta/2, \delta/2)$, the first leaves $(0, \delta, \delta/2)$ and the second restores the initial balances. Repeat $\theta/(2\delta)$ times. Paying each supplier $\delta/10$ from the demander gives voluntary, balanced compensation. Activities increase to the target, all prefixes are feasible, and the reserve is returned. The divisible frontier supplies the additional finite-N lower bound.

For the upper bound, take the attaining N-approval divisible schedule from Appendix B.2, at reserve $e_N = \theta/[12(N-1)]$. Round its cumulative activities down: $\widetilde{z}^t_k = \delta\lfloor z^t_k/\delta \rfloor$. Rounding preserves monotonicity, caps, and each approval's participant support. Initial and terminal allocations are unchanged because the target is grid-compatible. For $r^t = \widetilde{z}^t - z^t \in (-\delta, 0]^3$, the positive coefficients in each row of M_0 sum to at most four, so $M_0 r^t \geq -4\delta\mathbf{1}$. All rounded prefixes are therefore feasible with reserve $(e_N + 4\delta)\mathbf{1}$; their linear interpolations are feasible as well. The unchanged terminal target has $M_0 z_\theta = 0$, allowing full repayment.

The attaining schedule has minimum transaction surplus $\eta = \min_t(x_t - y_t/10) > 0$. Each rounded increment differs from its original by less than δ in each active coordinate. Its surplus is therefore at least $\eta - 11\delta/10 > 0$ whenever $\delta < 10\eta/11$. Paying each supplier its incremental cost gives balanced voluntary compensation in at most N approvals. The upper bound and the stated limit follow. *Q.E.D.*

Quantity loss and costs. For $0 < \delta \leq 1/2$, the largest grid-feasible repayable quantity under the unit caps is $\theta_\delta = 2\delta\lfloor 1/(2\delta) \rfloor$; its gross welfare loss satisfies $0 \leq \frac{9}{10}(1 - \theta_\delta) < \frac{9}{5}\delta$. Thus the grid-induced quantity loss and minimum seed vanish with δ. The minimum-seed construction uses θ_δ/δ approvals and need not minimize cost when approvals are priced. For a fixed cap, the rounding construction instead implements a compatible divisible target with at most $4\rho\delta$ additional reserve cost and no additional approvals. Coarse units and finite approval budgets therefore impose distinct constraints on implementation.

APPENDIX M: WATER TRADING: DATA, ORDERING, AND ACCOUNT CONSTRAINTS

M.1. *Data and reconciliation*

The application uses the Victorian Water Register's published reports for the Goulburn to Murray trade openings in October and December 2025 (DEECA, 2025a,b). The extraction preserves the processing priority, submitted volume, decision, and trading opportunity before and after each decision. Table I summarizes the 391 eligible applications. Quantities are represented as integer tenths of a megalitre, so all balance checks and counterfactual comparisons use exact arithmetic.

TABLE I

RECONCILIATION OF THE PUBLISHED PROCESSING LEDGERS

	October	December
Eligible applications	203	188
Approved applications	98	117
Requested volume (ML)	70,222.0	68,123.1
Opportunity before processing (ML)	31,215.1	37,647.4
External change during processing (ML)	0.0	−6.2
Approved volume (ML)	31,209.8	37,639.9
Opportunity after processing (ML)	5.3	1.3

October applications were lodged on 15 October and processed on 16 October. The opening opportunity of $31,523.4$ ML was reduced by 326.3 ML of tagged use and increased by 18.0 ML of back trade before the eligible batch was processed. Its initial opportunity was therefore $31,215.1$ ML. The analysis includes the complete eligible submission batch in Table 5 of the report. It excludes the two ineligible applications and the separately reported 5.3 ML application lodged later. Every ordering comparison uses these same 203 requests.

Recorded priorities and receipt identifiers are unique. Receipt identifiers were assigned in arrival order, which supplies the chronological counterfactual. The recorded sequence is reproduced exactly: each of the 203 decisions equals the comparison of requested volume with available opportunity, and every balance transition equals the approved debit. Rejected requests leave capacity unchanged and processing continues. Equal volumes in the smallest

first comparison are ordered by recorded priority. This convention selects one allocation without changing the maximum count.

M.2. *The count benchmark and its proof*

Fix eligible applications $\mathcal{I}$ with volumes $v_i > 0$, initial corridor opportunity $R \geq 0$, and eligibility held constant across orders. Each application executes in full or is rejected; none releases corridor capacity. For a permutation q, write

$$Y_i(q) = \mathbf{1}\left\{ v_i \leq R - \sum_{j \prec_q i} v_j Y_j(q) \right\}, \qquad A(q) = \sum_{i \in \mathcal{I}} Y_i(q).$$

Write $v_{(1)} \leq \cdots \leq v_{(n)}$, $S_0 = 0$, and $S_k = \sum_{j=1}^{k} v_{(j)}$.

PROPOSITION A.12—Application count and additional capacity: *The maximum count under sequential processing is*

$$k^*(R) = \max\{k \in \{0, \ldots, n\} : S_k \leq R\}.$$

Smallest first attains this count. Joint selection attains the same count, and every jointly feasible subset is sequentially feasible in every order. The minimum additional corridor capacity required for k applications is $(S_k - R)_+$. The maximizing order need not be unique.

PROOF: Every subset of k applications has volume at least S_k, so no feasible selection contains more than $k^*(R)$ applications. In increasing volume order, every prefix through $k^*(R)$ fits. The next request, if present, exceeds residual capacity, as do all subsequent requests. The rule therefore attains the upper bound. This selected subset is jointly feasible. Conversely, positive volumes imply that every prefix of a jointly feasible subset uses at most the subset's total volume. Every order of that subset is consequently feasible. Additional capacity c permits k requests exactly when $R + c \geq S_k$, giving the stated minimum. Any permutation of an optimal subset placed first attains the same count, so uniqueness is unnecessary. *Q.E.D.*

For October the exact certificate is

$$S_{159} = 31{,}193.0 \leq 31{,}215.1 < 31{,}693.0 = S_{160}.$$

Thus the scalar maximum is 159 applications. Executing 160 would require at least 477.9 ML of additional corridor capacity. This increment is consumed trading opportunity, distinct from the repayable temporary reserve in the main model. Twenty three requests have the cutoff volume of 500 ML, reinforcing the nonuniqueness of the maximizing allocation.

Relative to recorded priority, the selected allocation retains 79 approved applications, replaces 19, and admits 80 previously refused applications. The net increase is 61. Its executed volume is 16.8 ML lower. These selection margins distinguish a larger count from a Pareto improvement. All 203 applications underwent administrative processing; successful applications are also distinct from the costly approval steps priced by the main model.

The count objective gives each application unit weight. With gains g_i, the relevant allocation objective instead maximizes $\sum_i g_i x_i$ subject to $\sum_i v_i x_i \leq R$ and $x_i \in \{0, 1\}$. Smallest first solves that problem for equal positive gains, whereas heterogeneous gains generally require another selection. The October records classify 191 applications as related party trades and report a zero declared price for 174. Declared prices therefore do not identify gains from trade, and the empirical outcomes remain application counts and executed volumes. Applications also need not represent distinct traders.

M.3. *A conditional randomization benchmark*

The ACCC independently observed the October randomization procedure (ACCC, 2025). To characterize ordering variation within the fixed application menu, the replication draws 100,000 uniform permutations with NumPy PCG64 and seed 20251015. This stipulated benchmark yields a mean of 93.66423 approved applications, a standard deviation of 9.39806, and a median of 94. The central 95% interval runs from 75 to 112. The Monte Carlo standard error of the mean is 0.02972 applications.

The observed count of 98 is ordinary within this distribution. The comparison with smallest first reflects a systematic difference in selection rules, rather than an unusually unfavorable lottery draw. The simulated mean gap from 159 is 65.33577 applications. These quantities describe a conditional simulation distribution; they are not estimates of a causal population effect. A permanent policy change can change submissions and request sizes, including incentives to split applications. The exercise identifies the ordering margin for the

observed requests, while the institution's equity and allocation objectives govern the policy interpretation.

M.4. *Account restrictions and the connection to the theory*

December provides an additional ledger check. The opening release of $38,085.0$ ML was reduced by 437.6 ML of tagged use before processing, giving $37,647.4$ ML. A further 6.2 ML debit occurred between priorities 70 and 71. The application at priority 45 requested 62.3 ML and was refused while opportunity remained $23,824.0$ ML. Footnote 8 attributes the refusal to an insufficient seller account balance (DEECA, 2025b). Incorporating that eligibility restriction and the recorded external debit reproduces all 188 decisions and balance transitions, including the 1.3 ML final opportunity.

A seller's allocation account records private holdings. Its restriction establishes that aggregate corridor opportunity alone is insufficient for execution; it does not identify a separate public account of the main model. Recoverability would require a feasible transfer that credits the seller before processing its request. The reports do not provide the complete seller ledgers needed to establish such a path or to reconstruct eligibility under alternative October orders. Accordingly, 159 is the exact capacity benchmark conditional on fixed eligibility and an upper benchmark for a system with additional restrictions that only exclude transactions.

The application gives the theory a concrete institutional margin: almost identical aggregate utilization can support markedly different numbers of executed requests. In this scalar setting, optimal joint and sequential selection coincide. The main theory adds the economic content of separate accounts and productive releases, characterizing the cost of an admissible schedule when the authority chooses its allocation. The empirical application establishes the importance of selection order with positive existing capacity and identifies the account information needed to investigate those further mechanisms.

APPENDIX N: RELATIONSHIP TO EXISTING WORK

Table II relates the implementation-loss results to their mathematical and economic antecedents. The divided-list construction in Fraca and Haddad (2015, Lemma 13) provides an inverse-buffer upper bound. Our analysis adds a matching nonexpansion lower bound

and an exact frontier. The economic comparison allows each institution to choose its allocation subject to voluntary participation and balanced compensation. Standing capacity and productive releases then determine reserve borrowing and the maximum fee the authority will pay for joint authorization.

The precise continuous-net connection. Fix a finite list of voluntary bundles in the linear model. Represent each bundle u by a transition with

$$\mathrm{Pre}(u) = ([-Mu]_+, u), \qquad \mathrm{Post}(u) = ([Mu]_+, 0).$$

The state $(q, \bar{z} - z)$ records balances and remaining activity caps. Firing in amount $s \geq 0$ is equivalent to $q + sMu \geq 0$ and $z + su \leq \bar{z}$. Voluntariness and balanced compensation scale with u. Rational coefficients permit transition rescaling to integer arc weights. This represents a fixed list of atomic approvals, not every continuously chosen bilateral bundle by one fixed finite net. Splitting an approval into separate activity moves can change feasibility and the approval count.

TABLE II: Antecedents and scope of the quantitative results

Established result	Result and boundary here
Reachability and inverse bounds. Activation and divided execution (Fraca and Haddad, 2015, Lemma 13, Theorem 20).	Voluntary atomic approvals, repayment, a matching nonexpansion lower bound, and an exact frontier for M_h.
Amplification. Enabled resource-producing directions (Fraca and Haddad, 2015, Theorem 25).	A finite executable release and completion give sharp logarithmic order; this is a sufficient class, not a classification of all technologies.
Costs. Square-root inventory minimization (Baumol, 1952).	Theorem 6bounds losses after allocation choice: $\Theta(\sqrt{\tau})$ versus $\Theta(\tau \log(1/\tau))$ at fixed ρ.

Established result	Result and boundary here
Settlement welfare and financing. Credit constraints and intraday finance (Kahn and Roberds, 2001); gridlock, collateral, and delay (Kahn et al., 2003); queue incentives (Martin and McAndrews, 2008); planner comparisons (Atalay et al., 2010).	Productive releases change the cost of an authority-selected execution schedule and its willingness to pay for joint authorization.

Theorem 6establishes matching orders for optimized implementation losses at a blocked start. Theorem 9

extends the quantitative comparison to positive standing balances under its uniform execution conditions. The fee thresholds translate these losses into institutional choice after allocation optimization. The execution certificate provides a sufficient class for the productive bound, while Proposition 2supplies the exact constants for M_h.

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